

Dummit & Foote Notes

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0 Preliminaries

0.1 Basics

◆ **DEFINITION:** The **order** or **cardinality** of a set A is denoted as $|A|$. If A is finite, then $|A|$ is the number of elements in A .

◆ **DEFINITION:** The **Cartesian Product** of sets A and B is the collection of elements

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

◆ **DEFINITION:** A **function** from A to B is denoted as $f : A \rightarrow B$, and the value of f at a is denoted as $f(a)$.

- The **domain** of f is A , and the **codomain** of f is B .
- We write $f : a \mapsto b$ or $a \mapsto b$ to denote $f(a) = b$.
- We say f is **well-defined** when $f(a)$ is unambiguous for every $a \in A$.
- The **range**, or **image of A under f** is the subset of B given by

$$f(A) = \{b \in B \mid f(a) = b \text{ for some } a \in A\}$$

- The **preimage** or **inverse image** of some subset C of B is the subset of A given by

$$f^{-1}(C) = \{a \in A \mid f(a) \in C\}$$

- The **fiber of f over b** for some $b \in B$ is the preimage of b , i.e., $f^{-1}(b)$.

◆ **DEFINITION:** The **identity map** on a set A is the function $\text{id}_A : A \rightarrow A$ given by $\text{id}_A(a) = a$ for every $a \in A$.

◆ **DEFINITION:** Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. Then the **composite map** $g \circ f : A \rightarrow C$ is given by

$$(g \circ f)(a) = g(f(a)).$$

◆ **DEFINITION:** Let $f : A \rightarrow B$ be a function.

- f is **injective**/an **injection** if $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$.
- f is **surjective**/a **surjection** if for every $b \in B$ there exists $a \in A$ such that $f(a) = b$, i.e., $f(A) = B$.
- f is **bijective**/a **bijection** if it is both injective and surjective.
- A **left inverse** of f is a function $g : B \rightarrow A$ such that $g \circ f : A \rightarrow A$ is id_A .
- A **right inverse** of f is a function $h : B \rightarrow A$ such that $f \circ h : B \rightarrow B$ is id_B .

♠ **PROPOSITION 0.1: CHARACTERIZATION OF INJECTIVITY, SURJECTIVITY, AND BIJECTIONS**

Let $f : A \rightarrow B$ be a function.

- f is injective if and only if f has a left inverse.
- f is surjective if and only if f has a right inverse.
- f is a bijection if and only if there exists $g : B \rightarrow A$ such that $f \circ g$ is the identity map on B and $g \circ f$ the identity map on A .
- If A and B are finite sets where $|A| = |B|$, then the following are equivalent:
 - f is bijective,
 - f is injective,
 - f is surjective.

Proof.

- Suppose f is injective. Fix some $a_0 \in A$ such that $f(a_0) \in f(A)$, and let $g : B \rightarrow A$ be defined as follows:

$$g(b) = \begin{cases} a & a \in A \text{ such that } f(a) = b \\ a_0 & b \in B - f(A) \end{cases}$$

To verify g is a left-inverse is trivial by construction. The converse is trivial.

- Surjectivity of f implies nonempty fibers over every $b \in B$, hence there exists $a_b \in A$ such that $f(a_b) = b$. Define $h : B \rightarrow A$ by $h(b) = a_b$. This is clearly a right inverse of f . The converse is trivial.
- If f is bijective, there exists a unique $a \in A$ for every $b \in B$ such that $f(a) = b$. Define $g : B \rightarrow A$ by $g(b) = a$. It is clear that $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. The converse is trivial.

- (iv) (a) implies (b) trivially. If (b) is true, then distinct elements of A map to distinct elements of B . $|A| = |B|$ implies every element of B is mapped to, hence (c) is true. If (c) is true, then every element of B has nonempty fiber. Since f is well-defined, and $|A| = |B|$, then every fiber must be a singleton. ■

◆ **DEFINITION:** The **inverse** of a function $f : A \rightarrow B$ is the function g in [Proposition 0.1](#).

◆ **DEFINITION:** A **permutation** of a set A is a bijection from A to itself.

◆ **DEFINITION:** Let $A \subseteq B$ and $f : B \rightarrow C$. The **restriction** of f to A is the function $f|_A$.

◆ **DEFINITION:** Let $A \subseteq B$ and $g : A \rightarrow C$. The **extension** of g to B is a function $f : B \rightarrow C$ such that $f|_A = g$. Note that an extension of g to B may not be unique, and may not even exist.

◆ **DEFINITION:** Let A be nonempty.

- A **binary relation** on A is some $R \subseteq A \times A$. We write $a \sim b$ if $(a, b) \in R$.
- We say $\sim$ is:
 - ◊ **reflexive** if $a \sim a$ for every $a \in A$,
 - ◊ **symmetric** if $a \sim b$ implies $b \sim a$ for every $a, b \in A$,
 - ◊ **transitive** if $a \sim b$ and $b \sim c$ implies $a \sim c$ for every $a, b, c \in A$.
- A **equivalence relation** on A is a binary relation that is reflexive, symmetric, and transitive.
- Let $\sim$ be an equivalence relation on A . The **equivalence class** of $a \in A$ is the set $[a] = \{b \in A \mid a \sim b\}$. Elements of $[a]$ are **equivalent** to a and are called **representatives** of $[a]$.
- Let I be some indexing set. A **partition** of A is a collection of nonempty subsets $\{A_i \mid i \in I\}$ such that
 - ◊ $A = \bigcup_{i \in I} A_i$,
 - ◊ $A_i \cap A_j = \emptyset$ for every $i, j \in I$ with $i \neq j$.

♠ **PROPOSITION 0.2: FUNDAMENTAL THEOREM OF EQUIVALENCE RELATIONS**

Let A be a nonempty set.

- (i) If $\sim$ is an equivalence relation on A , then the set of equivalence classes of $\sim$ form a partition of A .
- (ii) If $\{A_i \mid i \in I\}$ is a partition of A , then there is an equivalence class relation on A whose equivalence classes are the sets A_i for $i \in I$.

Proof.

- (i) Let B be the union of the set of equivalence classes of $\sim$. It is easy to see that $B = A$. To show that the equivalence classes are pairwise disjoint, it suffices to show that they must be equal if they are not disjoint. Let $[a], [b]$ have nontrivial intersection, and let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. Then $a \sim b$, hence $[a] = [b]$.
- (ii) Define a relation on A as follows: $a \sim b$ if and only if $a, b \in A_i$ for some $i \in I$. It is easy to verify that $\sim$ is an equivalence relation, and that the equivalence classes of $\sim$ are precisely the sets A_i for $i \in I$. ■

0.2 Properties of the Integers

◆ **DEFINITION: Well Ordering of $\mathbb{Z}$:** If $A \subseteq \mathbb{Z}^+$ is nonempty, there exists $m \in A$ such that $m \leq a$ for every $a \in A$ called the **minimal element** of A .

◆ **DEFINITION:** For $a, b \in \mathbb{Z}$ with $a \neq 0$, then a **divides** b , denoted as $a \mid b$, if there exists $k \in \mathbb{Z}$ such that $b = ak$. If $a \nmid b$, we write $a \nmid b$.

◆ **DEFINITION:** For nonzero $a, b \in \mathbb{Z}$, the **greatest common divisor** of a and b , denoted as (a, b) , is a unique positive integer d such that

- $d \mid a$ and $d \mid b$,
- if $c \in \mathbb{Z}$ such that $c \mid a$ and $c \mid b$, then $c \mid d$.

If $(a, b) = 1$, we say a and b are **relatively prime**.

◆ **DEFINITION:** For $a, b \in \mathbb{Z}$ with $a \neq 0$, the **least common multiple** of a and b , denoted as $\text{lcm}(a, b)$, is a unique positive integer ℓ such that

- $a \mid \ell$ and $b \mid \ell$,
- if $c \in \mathbb{Z}$ such that $a \mid c$ and $b \mid c$, then $\ell \mid c$.

Note that $d\ell = ab$.

◆ **DEFINITION:** The **Division Algorithm** states that for $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^+$, there exist unique $q, r \in \mathbb{Z}$ such that

$$a = bq + r, \quad 0 \leq r < b$$

The integer q is called the **quotient** and r the **remainder** of a upon division by b .

◆ **DEFINITION:** The **Euclidean Algorithm** is a method for computing the greatest common divisor of two integers a and b by repeatedly applying the Division Algorithm. More precisely, we have

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \\ r_{n-2} &= r_{n-1}q_n + r_n \\ r_{n-1} &= r_nq_{n+1} \end{aligned}$$

where $q_i \in \mathbb{Z}$ and $r_i \in \mathbb{Z}$ for every i . The algorithm terminates when we obtain a remainder of 0, and the last nonzero remainder is the greatest common divisor of a and b . We may backtrack the Euclidean Algorithm to find a **$\mathbb{Z}$ -linear combination** of a and b that equals (a, b) , i.e., we can find $x, y \in \mathbb{Z}$ such that

$$ax + by = (a, b).$$

✓ **EXAMPLE:** Let $a = 63$ and $b = 39$. Applying the Euclidean Algorithm, we have

$$\begin{aligned} 63 &= 39(1) + 24 \\ 39 &= 24(1) + 15 \\ 24 &= 15(1) + 9 \\ 15 &= 9(1) + 6 \\ 9 &= 6(1) + 3 \\ 6 &= 3(2) \end{aligned}$$

so that $(63, 39) = 3$. To find the $\mathbb{Z}$ -linear combination of 63 and 39 that equals 3, we can backtrack the Euclidean Algorithm as follows: Observe that $3 = 9 - 6(1)$. Then $3 = 9 - (15 - 9(1))(1) = 2(9) - 15$. Continuing in this way, we obtain $3 = 5(63) - 8(39)$.

◆ **DEFINITION:** A $p \in \mathbb{Z}^+$ is **prime** if its only positive divisors are 1 and p . Otherwise, p is **composite**. Moreover, if $p \mid ab$ for some $a, b \in \mathbb{Z}$, then $p \mid a$ or $p \mid b$.

◆ **DEFINITION:** The **Fundamental Theorem of Arithmetic** states that every integer $n > 1$ can be written as a product of primes, and this factorization is unique up to the order of the factors. More precisely, suppose

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k} = q_1^{\beta_1} q_2^{\beta_2} \cdots q_m^{\beta_m},$$

where p_i and q_j are primes and $\alpha_i, \beta_j \in \mathbb{Z}^+$. Then $k = m$ and, after reordering, $p_i = q_i$ and $\alpha_i = \beta_i$ for all i .
For $a, b \in \mathbb{Z}$ with prime factorizations

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}, \quad b = p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k},$$

where p_i are the primes that divide a or b , and $\alpha_i, \beta_i \in \mathbb{Z}^+$ with $\alpha_i = 0$ if $p_i \nmid a$ and $\beta_i = 0$ if $p_i \nmid b$, then

$$(a, b) = p_1^{\min(\alpha_1, \beta_1)} p_2^{\min(\alpha_2, \beta_2)} \cdots p_k^{\min(\alpha_k, \beta_k)}, \quad \text{lcm}(a, b) = p_1^{\max(\alpha_1, \beta_1)} p_2^{\max(\alpha_2, \beta_2)} \cdots p_k^{\max(\alpha_k, \beta_k)}.$$

♦ **DEFINITION:** The **Euler φ -Function** is defined as such: for $n \in \mathbb{Z}^+$, then $\varphi(n)$ is the number of positive integers less than or equal to n that are relatively prime to n . We have two main properties:

- If p is prime and $\alpha \in \mathbb{Z}^+$, then $\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$.
- If $m, n \in \mathbb{Z}^+$ are relatively prime, then $\varphi(mn) = \varphi(m)\varphi(n)$.

0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n

♦ **DEFINITION:** The **congruence relation** modulo n is the equivalence relation on $\mathbb{Z}$ defined as follows: for $a, b \in \mathbb{Z}$, we say a is **congruent** to b modulo n , denoted as $a \equiv b \pmod{n}$, if $n \mid (a - b)$. The **congruence class**, or **residue class** of a modulo n is the equivalence class of a under the congruence relation, denoted as $\bar{a}$. The set of all congruence classes modulo n is denoted as $\mathbb{Z}/n\mathbb{Z}$, also called the **integers modulo n** .

♦ **DEFINITION: Modular arithmetic** is the arithmetic of $\mathbb{Z}/n\mathbb{Z}$ under the operations of addition and multiplication defined as follows: for $\bar{a}, \bar{b} \in \mathbb{Z}/n\mathbb{Z}$, we define

$$\bar{a} + \bar{b} = \overline{a + b}, \quad \bar{a} \cdot \bar{b} = \overline{ab}$$

♥ **THEOREM 0.3: MODULAR ARITHMETIC IS WELL-DEFINED**

The operations of addition and multiplication on $\mathbb{Z}/n\mathbb{Z}$ are well-defined. More precisely, if $a, x \in \mathbb{Z}$ and $b, y \in \mathbb{Z}$ where $\bar{a} = \bar{x}$ and $\bar{b} = \bar{y}$, then $\overline{a + b} = \overline{x + y}$ and $\overline{ab} = \overline{xy}$.

Proof. If $\bar{a} = \bar{x}$, then $a - x = sn$ for some $s \in \mathbb{Z}$. Similarly, if $\bar{b} = \bar{y}$, then $b - y = tn$ for some $t \in \mathbb{Z}$. Then

$$(a + b) - (x + y) = (a - x) + (b - y) = sn + tn = (s + t)n$$

and

$$(ab) - (xy) = a(b - y) + (a - x)y = a(tn) + (sn)y = (at + sy)n$$

which shows that $\overline{a + b} = \overline{x + y}$ and $\overline{ab} = \overline{xy}$. ■

♠ **PROPOSITION 0.4:** $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \mid (a, n) = 1\}$

Proof. In exercises. ■

1 Introduction to Groups

1.1 Basic Axioms and Examples

◆ **DEFINITION:** A **binary operation** on a set G is a function $\star : G \times G \rightarrow G$. For $a, b \in G$, we write $a \star b$ instead of $\star(a, b)$.

◆ **DEFINITION:** A binary operation $\star$ on G is **associative** if $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$.

◆ **DEFINITION:** Let $\star$ be a binary operation on G . Then $a, b \in G$ are said to **commute** if $a \star b = b \star a$. If every pair of elements in G commute, then $\star$ is said to be **commutative**.

◆ **DEFINITION:** Let $H \subseteq G$ and $\star$ be a binary operation on G . Then H is **closed** under $\star$ if $a \star b \in H$ for all $a, b \in H$. Note that if $\star$ is associative or commutative on G , then it is also associative or commutative on H .

◆ **DEFINITION:** A **group** is an ordered pair $(G, \star)$ with set G and binary operation $\star$ such that the following axioms hold:

- (i) $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$, i.e., $\star$ is associative,
- (ii) there exists $e \in G$, called the **identity** of G , such that $e \star a = a \star e = a$ for all $a \in G$, and
- (iii) for each $a \in G$, there exists $a^{-1} \in G$ (called the **inverse** of a) such that $a \star a^{-1} = a^{-1} \star a = e$.

Moreover, we say $(G, \star)$ is **abelian** if $\star$ is commutative.

◆ **DEFINITION:** Let $(G, \star)$ and $(H, \diamond)$ be groups. The **direct product** of G and H is the group $(G \times H, *)$ where the elements are from the Cartesian product and $*$ is defined componentwise:

$$(g_1, h_1) * (g_2, h_2) = (g_1 \star g_2, h_1 \diamond h_2) \quad \text{for all } g_1, g_2 \in G \text{ and } h_1, h_2 \in H.$$

◆ PROPOSITION 1.1: GROUP PROPERTIES

Let $(G, \star)$ be a group with $g, h \in G$. Then the following hold:

- (i) The identity of G is unique,
- (ii) the inverse of g is unique,
- (iii) $(g^{-1})^{-1} = g$,
- (iv) $(g \star h)^{-1} = h^{-1} \star g^{-1}$,
- (v) for $g_1, g_2, \dots, g_n \in G$, the expression $g_1 \star g_2 \star \dots \star g_n$ is unambiguous.

Proof.

- (i) Let e and e' be two identities of G . Then $e = e \star e' = e'$.
- (ii) Let h and h' be two inverses of g . Then $h = h \star e = h \star (g \star h') = (h \star g) \star h' = e \star h' = h'$.
- (iii) Note that $g \star g^{-1} = e$ so that g^{-1} is an inverse of g^{-1} . By the uniqueness of inverses, we have $(g^{-1})^{-1} = g$.
- (iv) We have $(g \star h) \star (g \star h)^{-1} = e$. Multiplying on the left by g^{-1} then h^{-1} proves the result.
- (v) We proceed by induction. Note that $n = 1, 2, 3$ are trivial. Assume for every $k < n$ that k elements $g_1, g_2, \dots, g_k$ can be unambiguously expressed as $g_1 \star g_2 \star \dots \star g_k$. Then for n elements, we can split the product $g_1 \star g_2 \star \dots \star g_n$ into two products of k and $n - k$ elements for some $1 \leq k < n$, and by the inductive hypothesis, these two products are unambiguous. Since $\star$ is associative, the result follows. ■

◆ PROPOSITION 1.2: CANCELLATION LAWS

Let G be a group with $g, h \in G$. Then for any $a, b \in G$, the equations $ag = h$ and $gb = h$ have unique solutions for a and b , respectively. In particular,

- (i) if $ag = ah$, then $g = h$, and
- (ii) if $ga = ha$, then $g = h$.

Proof. We have $a = hg^{-1}$ and $b = g^{-1}h$, and uniqueness follows from uniqueness of inverses. (i) is satisfied by left-multiplying both sides by a^{-1} , and (ii) is satisfied by right-multiplying both sides by a^{-1} . ■

◆ **DEFINITION:** Let G be a group with $g \in G$. The **order** of g , denoted by $|g|$, is the smallest positive integer n such that $g^n = 1$. If no such n exists, then we say g has infinite order.

◆ **DEFINITION:** Let $G = \{g_1, g_2, \dots, g_n\}$ be a group with $g_1 = 1$. The **multiplication table**, or **group table**, of G is the $n \times n$ matrix whose (i, j) -entry is $g_i g_j$ for $1 \leq i, j \leq n$.

1.2 Dihedral Groups

◆ **DEFINITION:** Let $n \geq 3$ be an integer. The **dihedral group** of order $2n$, denoted by D_{2n} , is the group of symmetries of a regular n -gon. Two elements of D_{2n} are the rotation r by $2\pi/n$ radians and the reflection s across a line of symmetry.

► **NOTE:** The following are claims about D_{2n} :

- (i) $1, r, r^2, \dots, r^{n-1}$ are all distinct, and $r^n = 1$ so that $|r| = n$.
- (ii) $|s| = 2$.
- (iii) $s \neq r^i$ for any i .
- (iv) $sr^i \neq sr^j$ for all $0 \leq i, j \leq n-1$ with $i \neq j$, so

$$D_{2n} = \{1, r, r^2, \dots, r^{n-1}, s, sr, sr^2, \dots, sr^{n-1}\}.$$

(v) $rs = sr^{-1}$.

(vi) $r^i s = sr^{-i}$ for all $0 \leq i \leq n$.

Proof.

- (i) Clearly, no two rotations by different angles are the same. Moreover, a $2\pi k$ rotation is the identity if and only if $n \mid k$, hence $|r| = n$.
- (ii) Clear.
- (iii) Reflections do not preserve orientation while rotations do, so s cannot be a power of r .
- (iv) If (iv) were true, then it would imply $r^i = r^j$ for some $i \neq j$, which contradicts (i).
- (v) We consider two cases:
 - (a) If n is odd, then $s(1) = 1$ and $s(i) = n + 2 - i$.
 - (b) If n is even, then $s(1) = 1$ and $s(n/2 + 1) = n/2 + 1$. For other i , then $s(i) = n + 2 - i$.
 For r , we have $r(n) = 1$ and $r(i) = i + 1$ for $1 \leq i \leq n-1$. Then $r(s(1)) = r(1) = 2$ and $r(s(2)) = r(n) = 1$. On the other hand, $s(r^{-1}(1)) = s(n) = 1$ and $s(r^{-1}(2)) = s(1) = 1$. Hence rs and sr^{-1} have the same effect on vertices 1 and 2, and they must be the same permutation.
- (vi) By (v), $i = 1$ holds true. Suppose it is true for $i \in \mathbb{Z}^+$. Then for $i + 1$, we have $r^{i+1}s = r(r^i s) = r(sr^{-i}) = (sr^{-1})r^{-i} = sr^{-(i+1)}$, and the result follows by induction. ■

◆ **DEFINITION:** A set of **generators** of a group G is a subset $S \subseteq G$ such that every element of G can be expressed as a finite product of elements of S and their inverses. We denote this as $G = \langle S \rangle$. We also say S **generates** G .

◆ **DEFINITION:** Equations that are satisfied by the generators of a group are called **relations**.

◆ **DEFINITION:** Let $G = \langle S \rangle$, and let $R_1, \dots, R_m$ be equations satisfied by elements of $S \cup \{1\}$. Then the generators and relations form a **presentation** of G , denoted by

$$G = \langle S \mid R_1, \dots, R_m \rangle.$$

A presentation for D_{2n} is given by

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

✓ **EXAMPLE:** The group X_{2n} with presentation

$$X_{2n} = \langle x, y \mid x^n = y^2 = 1, xy = yx^2 \rangle$$

has order at most 6 via the chain of equalities

$$x = xy^2 = (xy)y = (yx^2)y = (yx)(xy) = (yx)(yx^2) = y(xy)x^2 = y(yx^2)x^2 = x^4$$

✓ **EXAMPLE:** The group Y with presentation

$$Y = \langle u, v \mid u^4 = v^3 = 1, uv = v^2u^2 \rangle$$

is trivial.

1.3 Symmetric Groups

♦ **DEFINITION:** Let Ω be a nonempty set, and let S_Ω be the set of all permutations of Ω . Then $(S_\Omega, \circ)$ is a group under composition of functions, called the **symmetric group on Ω** .

♦ **DEFINITION:** Let $\Omega = \{1, 2, \dots, n\}$. Then S_Ω is denoted by S_n , and we say S_n is the **symmetric group of degree n** .

♦ **DEFINITION:** A **cycle** is a string of integers that represents a permutation in S_n that cyclically permutes the integers in the cycle and fixes all other integers. The number of integers in a cycle is its **length**, and if there are ℓ integers in the cycle, we call it an ℓ -**cycle**. We call two cycles with no common integers **disjoint cycles**.

♦ **DEFINITION:** The **cycle decomposition** of some $\sigma \in S_n$ is the product of disjoint cycles that represents σ . Note that the cycle decomposition of σ is unique up to the order of the cycles and the starting point of each cycle.

♦ **DEFINITION:** The **cycle decomposition algorithm** is a set of steps that can be used to find the cycle decomposition of some $\sigma \in S_n$:

- (i) Start with the smallest integer from $\{1, 2, \dots, n\}$ that has not appeared in any previous cycle. Typically, we start with $a = 1$ if no other cycle has been formed yet.
- (ii) Apply σ to a to get $\sigma(a)$. If $\sigma(a) = a$, then the cycle is (a) , and we can move on to the next smallest integer that has not appeared in any previous cycle. Otherwise, we continue applying σ to the result until we get back to a , and the cycle is formed by the integers in the order they were generated.
- (iii) Repeat the above steps until all integers have appeared in some cycle.
- (iv) Remove all single cycles of the form (a) since they represent the identity permutation on a .

✓ **EXAMPLE:** Let $n = 13$ and define $\sigma \in S_{13}$ as

$$\begin{aligned} \sigma(1) = 12, \quad \sigma(2) = 13, \quad \sigma(3) = 3, \quad \sigma(4) = 1, \quad \sigma(5) = 11, \quad \sigma(6) = 9, \quad \sigma(7) = 5, \\ \sigma(8) = 10, \quad \sigma(9) = 6, \quad \sigma(10) = 4, \quad \sigma(11) = 7, \quad \sigma(12) = 8, \quad \sigma(13) = 2 \end{aligned}$$

- Starting the cycle, begin with $a = 1$. Since $\sigma(1) = 12 \neq 1$, then $\sigma(12) = 8, \sigma(8) = 10, \sigma(10) = 4$, and $\sigma(4) = 1$. The first cycle is then $(1 \ 12 \ 8 \ 10 \ 4)$.
- The next smallest number is 2, so $\sigma(2) = 13$ and $\sigma(13) = 2$. Then the second cycle is $(2 \ 13)$.
- The next smallest number is 3, so $\sigma(3) = 3$, and the third cycle is (3) .
- The next smallest is 5, so $\sigma(5) = 11, \sigma(11) = 7$, and $\sigma(7) = 5$. Then the fourth cycle is $(5 \ 11 \ 7)$.
- The next smallest is 6, so $\sigma(6) = 9$ and $\sigma(9) = 6$. Then the fifth cycle is $(6 \ 9)$.
- Remove the single cycle (3) .

Following this, we have $\sigma = (1 \ 12 \ 8 \ 10 \ 4)(2 \ 13)(5 \ 11 \ 7)(6 \ 9)$.

► **NOTE:** S_n is a non-abelian group for all $n \geq 3$ since the cycles $(1 \ 2)$ and $(2 \ 3)$ do not commute.

► **NOTE:** Disjoint cycles permute only the integers in the cycle, hence they commute.

► **NOTE:** The order of an ℓ -cycle is simply ℓ . The order of a cycle with k disjoint cycles of lengths $\ell_1, \ell_2, \dots, \ell_k$ is $\text{lcm}(\ell_1, \ell_2, \dots, \ell_k)$.

1.4 Matrix Groups

◆ **DEFINITION:** A **field** is a set F with two binary operations $+$ and $\cdot$ such that $(F, +)$ is abelian with identity 0, and $(F - \{0\}, \cdot)$ is abelian with identity 1. Moreover, the following distributive law holds:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \text{for all } a, b, c \in F.$$

Moreover, we denote $F^\times = F - \{0\}$.

◆ **DEFINITION:** Let F be a field. The **general linear group of degree n** is the set:

$$\text{GL}_n(F) = \{A \mid A \text{ is an } n \times n \text{ matrix with entries from } F \text{ and } \det(A) \neq 0\}$$

with the binary operation of matrix multiplication.

► **NOTE:** Some recorded results for fields, which are proven later:

- if F is a field with $|F| < \infty$, then $|F| = p^k$ for prime p and $k \in \mathbb{Z}^+$, and
- if $|F| = p < \infty$, then $|\text{GL}_n(F)| = (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{n-1})$.

1.5 The Quaternion Group

◆ **DEFINITION:** The **quaternion group**, denoted by Q_8 , is defined to be the set

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}.$$

It has operation $\cdot$ defined by the following rules:

- (i) $1 \cdot q = q \cdot 1 = q$ for all $q \in Q_8$,
- (ii) $(-1) \cdot (-1) = 1, (-1) \cdot q = q \cdot (-1) = -q$ for all $q \in Q_8$,
- (iii) $i^2 = j^2 = k^2 = -1$,
- (iv) $ij = k, jk = i, ki = j$, and
- (v) $ji = -k, kj = -i, ik = -j$.

This is a non-abelian group of order 8.

1.6 Homomorphisms and Isomorphisms

◆ **DEFINITION:** Let $(G, \star)$ and $(H, \diamond)$ be groups. A **homomorphism** is a map $\varphi : G \rightarrow H$ such that

$$\varphi(g \star h) = \varphi(g) \diamond \varphi(h), \quad \text{for all } g, h \in G.$$

φ is an **isomorphism** if φ is a bijection. If there exists an isomorphism from G to H , then we say G and H are **isomorphic**, denoted by $G \cong H$. Moreover, for two isomorphic groups G and H , it is to be deduced that

- (i) $|G| = |H|$,
- (ii) G is abelian if and only if H is abelian, and
- (iii) for all $g \in G$, then $|g| = |\varphi(g)|$.

✓ **EXAMPLE:** Let Γ and Σ be two nonempty sets. Then $S_\Gamma \cong S_\Sigma$ if and only if $|\Gamma| = |\Sigma|$. A sketch is as follows:

- (i) Let $\theta : \Gamma \rightarrow \Sigma$ be some bijection. Then the map

$$\varphi : S_\Gamma \rightarrow S_\Sigma, \quad \varphi(\sigma) = \theta \circ \sigma \circ \theta^{-1}$$

is an isomorphism.

- (ii) If $S_\Gamma \cong S_\Sigma$, we note that for finite Γ and Σ where $|\Gamma| = m$ and $|\Sigma| = n$, then $m! = n!$ implies $m = n$.

► **NOTE:** Let $|G| = n < \infty$ with some presentation, and let $S = \{s_1, s_2, \dots, s_k\}$ generate G . Let H be another group with $\{r_1, \dots, r_k\} \subseteq H$.

- If the r_i satisfy the same relations as the s_i , then there is a unique homomorphism $\varphi : G \rightarrow H$ such that $\varphi(s_i) = r_i$ for all i .
- If $H = \langle r_1, \dots, r_k \rangle$ and the r_i satisfy the same relations as the s_i , then φ is surjective.
- If H is finite and $|H| = |G|$, then φ is injective, hence an isomorphism.

1.7 Group Actions

◆ **DEFINITION:** A **group action** of a group G on a set A is a map $G \times A \rightarrow A$, written as $g \cdot a$ for $g \in G$ and $a \in A$, such that

- (i) $g \cdot (h \cdot a) = (gh) \cdot a$ for all $g, h \in G$ and $a \in A$, and
- (ii) $1 \cdot a = a$ for all $a \in A$.

◆ **DEFINITION:** Let a group G act on a set A . For fixed $g \in G$, there exists a permutation σ_g on A such that

$$\sigma_g(a) = g \cdot a.$$

The map $g \mapsto \sigma_g$ is a homomorphism from G to S_A , called the **permutation representation** of G on A . If instead we have a homomorphism $\varphi : G \rightarrow S_A$, then we can define a group action of G on A by $g \cdot a = \varphi(g)(a)$ for all $g \in G$ and $a \in A$.

◆ **DEFINITION:** Let G act on A .

- The **trivial action** of G on A is defined by $ga = a$ for all $g \in G$ and $a \in A$. Moreover, the associated permutation representation is the homomorphism $\varphi : G \rightarrow S_A$ defined by $\varphi(g) = \text{id}$ for all $g \in G$.
- If distinct elements of G induce distinct permutations on A , then the action is **faithful**. In this case, the associated permutation representation is injective.
- The **kernel** of the action is the set $\{g \in G \mid ga = a \text{ for all } a \in A\}$.

◆ **DEFINITION:** Let G act on itself. Then the **left regular action** of G on itself is defined by $g \cdot a = ga$ for all $g, a \in G$. Note that this action is faithful, since $ga = ha$ for all $a \in G$ implies $g = h$.

2 Subgroups

2.1 Definitions and Examples

◆ **DEFINITION:** Let G be a group with $H \subseteq G$. Then H is a **subgroup** of G if H is nonempty and closed under products and inverses. We denote this with $H \leq G$.

♠ **PROPOSITION 2.1: SUBGROUP CRITERION**

For some group G with $H \subseteq G$, then $H \leq G$ if and only if

- (i) H is nonempty, and
- (ii) for all $g, h \in H$, then $gh^{-1} \in H$.

If H is finite, then $H \leq G$ if and only if H is nonempty and closed under products.

Proof. If $H \leq G$, then $1 \in H$. Moreover, H contains its inverses and is closed, hence both conditions are satisfied.

Conversely, suppose H satisfies (i) and (ii). Nonemptiness implies that there exists $h \in H$. Then $hh^{-1} = 1 \in H$. Moreover, we have $h^{-1}1 = h^{-1} \in H$ so that H contains inverses. Finally, for any $g, h \in H$, then $gh = g(h^{-1})^{-1} \in H$ so that H is closed under products. Hence, $H \leq G$.

If H is finite and closed under products, let $h \in H$. Then there are only finitely many distinct elements in the list $h, h^2, h^3, \dots$. There must exist $s, t \in \mathbb{Z}$ such that $h^s = h^t$ where $s < t$. Then $h^{t-s} = 1$ so that $h^{-1} = h^{t-s-1} \in H$. Hence, H contains inverses, so $H \leq G$. ■

2.2 Centralizers and Normalizers, Stabilizers and Kernels

Let G be a group with nonempty $S \subseteq G$.

◆ **DEFINITION:** The **centralizer** of S in G is the set

$$C_G(S) = \{g \in G \mid gsg^{-1} = s \text{ for all } s \in S\}.$$

◆ **DEFINITION:** The **normalizer** of S in G is the set

$$N_G(S) = \{g \in G \mid gSg^{-1} = S\},$$

where $gSg^{-1} = \{gsg^{-1} \mid s \in S\}$.

◆ **DEFINITION:** The **center** of G is the set

$$Z(G) = \{g \in G \mid gx = xg \text{ for all } x \in G\}.$$

► **NOTE:** To see $C_G(S) \leq G$, note that $1 \in C_G(S)$. For $x, y \in C_G(S)$, then

$$xy^{-1}s(xy^{-1})^{-1} = xy^{-1}syx^{-1} = xy^{-1}ysx^{-1} = xsx^{-1} = s.$$

Hence, $C_G(S) \leq G$. A similar argument shows that $N_G(S) \leq G$. Moreover, we have $gsg^{-1} = s$ for all $s \in G$ whenever $g \in C_G(S)$, hence $C_G(S) \leq N_G(S)$. Finally, $Z(G) = C_G(G)$ so that $Z(G) \leq G$.

✓ **EXAMPLE:** Let $G = S_3$ and $S = \{\text{id}, (1\ 2)\}$. We illustrate how to use Lagrange's Theorem to minimize calculations in showing that $C_G(S) = S$. Since $C_G(S) \leq G$, then $|C_G(S)|$ divides $|G| = 6$. Moreover, $C_G(S)$ contains the identity and $(1\ 2)$, so 2 divides $|C_G(S)|$. Hence, $|C_G(S)|$ is either 2 or 6. If $|C_G(S)| = 6$, then $C_G(S) = G$, but $(1\ 2\ 3) \notin C_G(S)$. Hence, $|C_G(S)| = 2$ so that $C_G(S) = S$.

Note that $\sigma \in N_G(S)$ if and only if $\sigma S \sigma^{-1} = S$. Since $\sigma \text{id} \sigma^{-1} = \text{id}$, it must be that $\sigma(1\ 2)\sigma^{-1} = (1\ 2)$. This occurs only when $\sigma \in C_G(S)$, so that $N_G(S) = C_G(S) = S$.

Since $(1\ 2) \notin Z(G)$, then $Z(G) = \{\text{id}\}$.

◆ **DEFINITION:** Let G act on A . The **stabilizer** of $a \in A$ is the set

$$G_a = \{g \in G \mid ga = a\}.$$

To see that $G_a \leq G$, note that $1a = a$. If $x \in G_a$, then $a = (x^{-1}x)a = x^{-1}a$. Finally, $x, y \in G_a$ implies that $xya = xa = a$. Hence, $G_a \leq G$. Moreover, the kernel of the action is also a subgroup of G .

◆ **DEFINITION:** Let G act on $\mathcal{P}(G)$, the set of subsets of G , by **conjugation**. More specifically, for $g \in G$ and $S \in \mathcal{P}(G)$, then

$$g : S \mapsto gSg^{-1} = \{gsg^{-1} \mid s \in S\}.$$

We see that $N_G(S)$ is the stabilizer of S under this action, hence $N_G(S) \leq G$.

If $N_G(S)$ acts on S by conjugation, then the kernel of this action is $C_G(S)$, hence $C_G(S) \leq N_G(S)$. Transitivity of subgroup relations gives $C_G(S) \leq N_G(S) \leq G$.

If G acts on itself by conjugation, then the kernel of this action is $Z(G)$, hence $Z(G) \leq G$.

2.3 Cyclic Groups and Cyclic Subgroups

◆ **DEFINITION:** A group G is cyclic if G is generated by a single element, i.e., there exists $g \in G$ such that $G = \{g^n \mid n \in \mathbb{Z}\}$. Such an element g is called a **generator** of G . We denote the cyclic group generated by g as $\langle g \rangle$. Note that cyclic groups are abelian.

♠ **PROPOSITION 2.2: CHARACTERIZATION OF ELEMENTS OF CYCLIC GROUPS**

Let $G = \langle g \rangle$. Then $|G| = |g|$. In particular,

- (i) If $|G| = n < \infty$, then $g^n = 1$, and the elements $1, g, g^2, \dots, g^{n-1}$ are distinct.
- (ii) If $|G| = \infty$, then $g^n \neq 1$ for any $n \neq 0$, and $g^s \neq g^t$ for any $s \neq t$ in $\mathbb{Z}$.

Proof.

- (i) Put $|g| = n$. Clearly, $1, g, g^2, \dots, g^{n-1}$ are distinct. If $g^k \in G$ for some $k \in \mathbb{Z}$, then by the Division Algorithm, there exists $q \in \mathbb{Z}$ and $0 \leq r < n$ such that $k = nq + r$. Then

$$g^k = g^{nq+r} = (g^n)^q g^r = 1^q g^r = g^r$$

so that g^k is one of the aforementioned elements. Hence, $G = \{1, g, g^2, \dots, g^{n-1}\}$.

- (ii) If $|g| = \infty$ and $g^s = g^t$ for some $s, t \in \mathbb{Z}$, then $g^{t-s} = 1$, contradicting that $|g| = \infty$. Hence, the elements g^n are all distinct, so that $|G| = \infty$. ■

♠ **PROPOSITION 2.3: DIVISIBILITY OF ORDERS**

Let G be a group with $g \in G$ and $s, t \in \mathbb{Z}$. If $g^s = g^t = 1$, then $g^d = 1$, where $d = (s, t)$. In particular, if $g^s = 1$ for some $s \in \mathbb{Z}$, then $|g|$ divides s .

Proof. The Euclidean Algorithm produces $x, y \in \mathbb{Z}$ such that $xs + yt = d$. Then

$$g^d = g^{xs+yt} = (g^s)^x (g^t)^y = 1^x 1^y = 1.$$

Hence, $g^d = 1$.

If $g^s = 1$ with $|g| = t$. Certainly, $t \mid s$ when $s = 0$. If $s \neq 0$, then $t < \infty$ since t is a nonzero power of g . Put $d = (s, t)$, hence $g^d = 1$ by the preceding paragraph. Since $0 < d \leq t$ and t is the smallest positive integer such that $g^t = 1$, then $d = t$, so that $t \mid s$. ■

♥ **THEOREM 2.4: CYCLIC GROUPS OF SAME ORDER ARE ISOMORPHIC**

Any two cyclic groups of the same order are isomorphic. In particular,

- (i) For $n \in \mathbb{Z}^+$ with $\langle g \rangle$ and $\langle h \rangle$ cyclic groups of order n , the map

$$\varphi : \langle g \rangle \rightarrow \langle h \rangle \quad \text{given by} \quad g^k \mapsto h^k$$

is well defined and an isomorphism.

- (ii) If $\langle g \rangle$ is an infinite cyclic group, then the map

$$\varphi : \mathbb{Z} \rightarrow \langle g \rangle \quad \text{given by} \quad k \mapsto g^k$$

is also well defined and an isomorphism.

Proof.

- (i) Suppose $g^a = g^b$ for $a, b \in \mathbb{Z}$. Then $g^{b-a} = 1$, hence $n \mid (b - a)$. Put $b = nt + a$ for some $t \in \mathbb{Z}$. Then

$$\varphi(g^b) = \varphi(g^{nt+a}) = h^{nt+a} = (h^n)^t h^a = 1^t h^a = h^a = \varphi(g^a).$$

Hence, φ is well defined. Moreover, the laws of exponents show that $\varphi(g^a g^b) = \varphi(g^a) \varphi(g^b)$. Finally, $h^k = \varphi(g^k)$, hence φ is surjective. **Proposition 0.1(iv)** finishes this.

- (ii) That φ is well defined and a homomorphism is clear. Surjectivity is also clear. **Proposition 2.2** shows that φ is injective, so **Proposition 0.1(iv)** finishes this. ■

► **NOTE:** We write Z_n for the cyclic group of order n . We have $Z_n = \langle z \rangle$ where $z^n = 1$. Note that $Z_n \cong \mathbb{Z}/n\mathbb{Z}$.

♠ **PROPOSITION 2.5: ORDERS OF ELEMENTS IN A CYCLIC GROUP**

Let G be a group with $g \in G$. Let $t \in \mathbb{Z}$ be nonzero.

- (i) If $|g| = \infty$, then $|g^t| = \infty$.
- (ii) If $|g| = n < \infty$, then $|g^t| = n/(t, n)$.
- (iii) If $|g| = n < \infty$ with $t \mid n$, then $|g^t| = n/t$.

Proof.

- (i) If $|g^t| = m < \infty$, then $1 = (g^t)^m = g^{tm}$, contradicting that $|g| = \infty$. Hence, $|g^t| = \infty$.
- (ii) Let $h = g^t$ and $d = (t, n)$. Note that $t = dx$ and $n = dy$ for $x, y \in \mathbb{Z}$ such that $(x, y) = 1$ and $y > 0$. Then

$$h^y = (g^t)^y = g^{ty} = g^{dxy} = g^{dnx} = (g^n)^{dx} = 1,$$

so that $|h|$ divides y by [Proposition 2.3](#). Moreover,

$$g^{t|h|} = h^{|h|} = 1,$$

so that $n \mid t|h|$ by [Proposition 2.3](#). Since $t = dx$ and $n = dy$, then $y \mid x|h|$. Since $(x, y) = 1$, then $y \mid |h|$. Hence, $|h| = y = n/d$.

- (iii) Note that $(t, n) = t$ since $t \mid n$. Hence, $|g^t| = n/t$ by the preceding part. ■

♠ **PROPOSITION 2.6: CYCLIC GROUP GENERATOR PROPERTIES**

Let $G = \langle g \rangle$.

- (i) If $|g| = \infty$, then $G = \langle g^a \rangle$ if and only if $a = \pm 1$.
- (ii) If $|g| = n < \infty$, then $G = \langle g^a \rangle$ if and only if $(a, n) = 1$.

Proof.

- (i) If $G = \langle g^a \rangle$, then $g = (g^a)^k$ for $k \in \mathbb{Z}$. This shows $ak = 1$, so $a = \pm 1$. The converse is clear.
- (ii) [Proposition 2.2](#) shows that $|\langle g^a \rangle| = |g^a|$. By [Proposition 2.5](#), we have $|g^a| = n/(a, n)$. Hence, $G = \langle g^a \rangle$ if and only if $n/(a, n) = n$, which occurs if and only if $(a, n) = 1$. ■

♥ **THEOREM 2.7: THE FUNDAMENTAL THEOREM OF CYCLIC GROUPS**

Let $G = \langle g \rangle$ be cyclic.

- (i) Every subgroup of G is cyclic: if $H \leq G$, then $H = 1$ or $H = \langle g^k \rangle$, where k is the smallest positive integer such that $g^k \in H$.
- (ii) If $|G| = \infty$, then for distinct $a, b \in \mathbb{Z}^+$, we have $\langle g^a \rangle \neq \langle g^b \rangle$. Moreover, for every $k \in \mathbb{Z}$, we have $\langle g^k \rangle = \langle g^{|k|} \rangle$. Hence, nontrivial subgroups of G correspond bijectively with $\mathbb{Z}^+$.
- (iii) If $|G| = n < \infty$, there exists a unique subgroup H of G with order x for every positive divisor x of n . This is the subgroup $\langle g^d \rangle$, where $d = n/x$. Moreover, for every $k \in \mathbb{Z}$, we have $\langle g^k \rangle = \langle g^{(n,k)} \rangle$, so that subgroups of G correspond bijectively with all positive divisors of n .

Proof.

- (i) Let $H \leq G$. $H = 1$ is trivial, so suppose $H \neq 1$. There exists $s \neq 0$ such that $g^s \in H$. For $s < 0$, we have $g^{-s} = (g^s)^{-1} \in H$. The set of positive integers t such that $g^t \in H$ is nonempty, so let d be the smallest such integer. Clearly, $\langle g^d \rangle \leq H$. For any $g^k \in H$, the Division Algorithm gives $k = dq + r$ for some $q \in \mathbb{Z}$ and $0 \leq r < d$. It is clear that $g^r \in H$ since $g^k \in H$ and $g^d \in H$. By minimality of d , we have $r = 0$, so that $g^k = g^{dq} \in \langle g^d \rangle$. Hence, $H \leq \langle g^d \rangle$, and we are done.
- (ii) If $\langle g^a \rangle = \langle g^b \rangle$ for distinct $a, b \in \mathbb{Z}^+$, then $g^b = (g^a)^u$ and $g^a = (g^b)^v$ for $u, v \in \mathbb{Z}$. Then $b = au = bvu$, or $1 = vu$. This occurs only when $u = v = 1$, contradicting that a and b are distinct. Hence, $\langle g^a \rangle \neq \langle g^b \rangle$ for distinct $a, b \in \mathbb{Z}^+$.

If $k \geq 0$, then $\langle g^k \rangle = \langle g^{|k|} \rangle$ is clear. If $k < 0$, then $k = -|k|$ so that $\langle g^k \rangle = \langle g^{-|k|} \rangle = \langle (g^{|k|})^{-1} \rangle = \langle g^{|k|} \rangle$ by [Proposition 2.6\(i\)](#). Hence, $\langle g^k \rangle = \langle g^{|k|} \rangle$ for all $k \in \mathbb{Z}$.

- (iii) Let $x \mid n$ and put $d = n/x$. [Proposition 2.5](#) asserts that $|\langle g^d \rangle| = n/d = x$, so there exists a subgroup of G with order x . For uniqueness, let H be another subgroup of G with order x . By part (1), we have $H = \langle g^b \rangle$ for some positive integer b such that $g^b \in H$. [Proposition 2.5](#) asserts that $|\langle g^b \rangle| = n/(n, b)$. Since $|\langle g^b \rangle| = x$, then $n/(n, b) = x$, so that $d = (n, b)$. Since $d \mid b$, then $g^b \in \langle g^d \rangle$, hence $\langle g^b \rangle \leq \langle g^d \rangle$. Since both subgroups have order x , then $\langle g^b \rangle = \langle g^d \rangle$, and we are done.

For any $k \in \mathbb{Z}$, we have $\langle g^k \rangle = \langle g^{(n,k)} \rangle$ by [Proposition 2.5](#) and [Proposition 2.2](#), so that subgroups of G correspond bijectively with all positive divisors of n . ■

2.4 Subgroups Generated by Subsets of a Group

♠ PROPOSITION 2.8: INTERSECTION OF SUBGROUPS IS A SUBGROUP

Let G be a group with a nonempty collection of subgroups $\mathcal{G}$. The intersection of all subgroups in $\mathcal{G}$ is a subgroup of G .

Proof. Follows from the fact that each $H \in \mathcal{G}$ is a subgroup of G . ■

◆ **DEFINITION:** Let G be a group with $S \subseteq G$. The **subgroup generated by S** is the set

$$\langle S \rangle = \bigcap_{\substack{S \subseteq H \\ H \leq G}} H.$$

We say that $\langle S \rangle$ is the minimal subgroup of G containing S . For subsets S, T of G , the subgroup generated by $S \cup T$ is denoted $\langle S, T \rangle$.

◆ **DEFINITION:** A finite product of elements of S is called a **word** in S . The set of all words is the set

$$\bar{S} = \{s_1^{\varepsilon_1} s_2^{\varepsilon_2} \cdots s_n^{\varepsilon_n} \mid n \in \mathbb{Z}^+ \cup \{0\}, s_i \in S, \varepsilon_i = \pm 1\}.$$

No assumptions of the countability of S are made, so that $\bar{S}$ may be uncountable.

♠ PROPOSITION 2.9: $\bar{S} = \langle S \rangle$

Proof. It is clear that $\bar{S} \leq G$. Since every $s \in S$ can be written as s^1 , then $S \subseteq \bar{S}$, or $\langle S \rangle \subseteq \bar{S}$. Conversely, $\bar{S}$ contains S , and since $\bar{S} \leq G$, then $\bar{S}$ is one of the subgroups in the intersection defining $\langle S \rangle$. Hence, $\bar{S} \subseteq \langle S \rangle$. ■

✓ **EXAMPLE:** If a group G is non-abelian, then subgroups of G generated by more than one element may be complicated. Let $G = \text{GL}_2(\mathbb{R})$ with

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix}$$

so that $A^2 = B^2 = I_2$. However,

$$AB = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix}, \quad (AB)^n = \begin{pmatrix} (1/2)^n & 0 \\ 0 & 2^n \end{pmatrix}$$

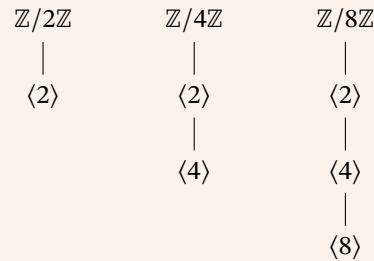
so that $\langle A, B \rangle$ is an infinite subgroup of $\text{GL}_2(\mathbb{R})$.

2.5 The Lattice of Subgroups of a Group

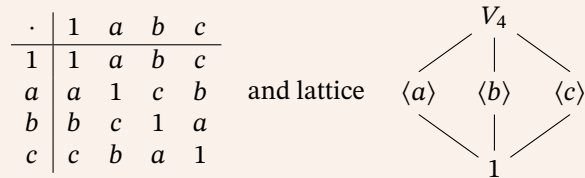
♦ **DEFINITION:** The **lattice of subgroups** of a group G is the graph whose vertices are the subgroups of G and whose edges are the subgroup relations. Moreover, for any pair of subgroups H and K , the **join** of H and K is the smallest subgroup of G containing both H and K , which is $\langle H, K \rangle$.

✓ **EXAMPLE:** Note that the following examples should only be taken as fact, with proofs of the subgroup lattices for *noncyclic* groups being seen later in the text.

- (i) For $Z_n \cong \mathbb{Z}/n\mathbb{Z}$, [Theorem 2.7](#) says that the subgroups of Z_n is the lattice of divisors of n . Specific examples are:

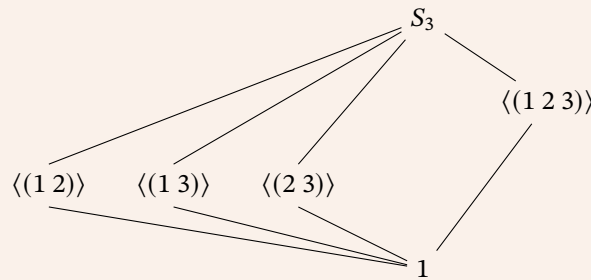


- (ii) The **Klein 4-group (Vierergruppe)**, V_4 , is the group of order 4 with each nonidentity element order 2

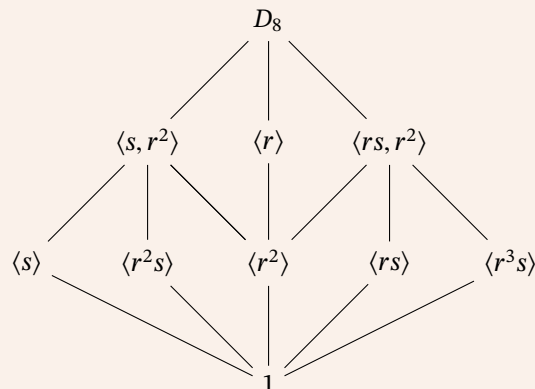


Note that V_4 is abelian that is not isomorphic to Z_4 since it does not contain elements of order 4.

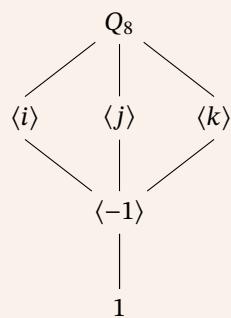
- (iii) S_3 is the lattice:



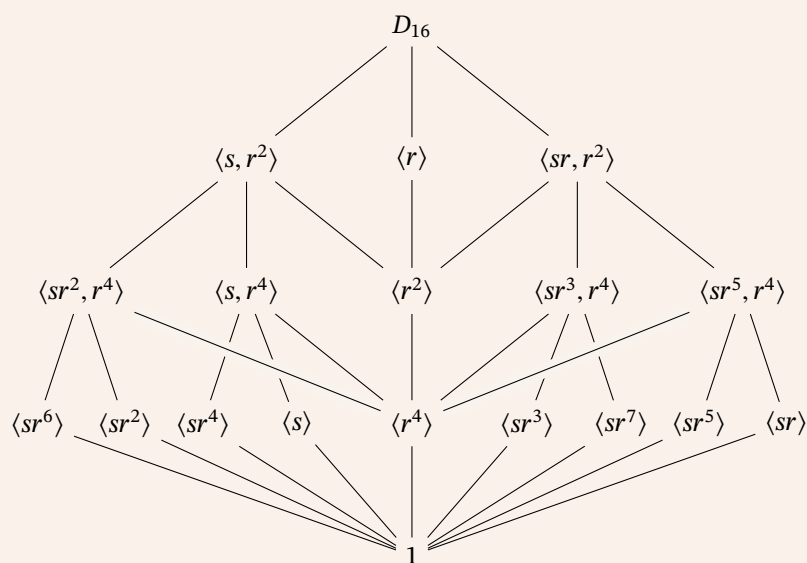
- (iv) D_8 is the lattice:



(v) Q_8 is the lattice:



(vi) D_{16} has a lattice that is not a planar graph, or is a graph drawn on a plane without lines crossing. A way of drawing it is



3 Quotient Groups and Homomorphisms

3.1 Definitions and Examples

◆ **DEFINITION:** Let $\varphi : G \rightarrow H$ be a homomorphism. The **kernel** of φ is the set

$$\ker \varphi = \{g \in G \mid \varphi(g) = 1\}$$

where 1_H is the identity of H . The **image** of φ is the set

$$\text{im } \varphi = \{\varphi(g) \mid g \in G\}$$

♠ **PROPOSITION 3.1: HOMOMORPHISM PROPERTIES**

Let $\varphi : G \rightarrow H$ be a group homomorphism. Then

- (i) $\varphi(1_G) = 1_H$, where 1_G and 1_H are the identities of G and H , respectively.
- (ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ for every $g \in G$.
- (iii) $\varphi(g^n) = \varphi(g)^n$ for every $n \in \mathbb{Z}$.
- (iv) $\ker \varphi \leq G$.
- (v) $\text{im } \varphi \leq H$.

Proof.

- (i) $\varphi(1_G) = \varphi(1_G 1_G) = \varphi(1_G)\varphi(1_G)$. By cancellation, $1_H = \varphi(1_G)$.
- (ii) $1_H = \varphi(1_G) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1})$ for all $g \in G$. Then $\varphi(g)^{-1} = \varphi(g^{-1})$.
- (iii) For $n = 0$, the result is (i). Using induction for $n \in \mathbb{Z}^+$, use (ii) for $n < 0$.
- (iv) Note $1_G \in \ker \varphi$. Suppose $x, y \in \ker \varphi$. Then $\varphi(xy^{-1}) = \varphi(x)\varphi(y^{-1}) = 1_H 1_H^{-1} = 1_H$ so that $xy^{-1} \in \ker \varphi$.
- (v) Note $1_H \in \text{im } \varphi$. Suppose $x, y \in \text{im } \varphi$. Then there exists $g, h \in G$ such that $\varphi(g) = x$ and $\varphi(h) = y$. Then $xy^{-1} = \varphi(g)\varphi(h^{-1}) = \varphi(gh^{-1})$. Since $gh^{-1} \in G$, then $xy^{-1} \in \text{im } \varphi$. ■

◆ **DEFINITION:** Let $\varphi : G \rightarrow H$ be a homomorphism with kernel K . The **quotient group** G/K is the group whose elements are the fibers of φ with operation as such: if $X = \varphi^{-1}(a)$ and $Y = \varphi^{-1}(b)$ for $a, b \in H$, then $XY = \varphi^{-1}(ab)$.

♠ **PROPOSITION 3.2: TRANSLATES OF A KERNEL**

Let $\varphi : G \rightarrow H$ be a homomorphism with kernel K . Let $X = \varphi^{-1}(a)$ for $X \in G/K$. Then

- (i) For any $x \in X$, $X = \{xk \mid k \in K\}$.
- (ii) For any $x \in X$, $X = \{kx \mid k \in K\}$.

Proof.

- (i) Let $x \in X$, hence $\varphi(x) = a$. Let $xK = \{xk \mid k \in K\}$. For any $k \in K$, then $xk \in X$, or $xK \subseteq X$. If $y \in X$, put $k = x^{-1}y$. Then $k \in K$ so that $y = xk \in xK$, hence $X \subseteq xK$.
- (ii) Define $Kx = \{kx \mid k \in K\}$. The proof follows similarly as (i), except we set $k = yx^{-1}$ so that $kx = y \in X$, hence $X \subseteq Kx$. ■

◆ **DEFINITION:** Let $N \leq G$ and $g \in G$. The **left coset** and **right coset** of N in G are the sets

$$gN = \{gn \mid n \in N\}, \quad Ng = \{ng \mid n \in N\}.$$

Elements of a coset are called **representatives**.

♥ **THEOREM 3.3: COSET MULTIPLICATION**

Let G be a group with K a kernel of some homomorphism from G to another group. The set of left cosets of K in G with operation

$$aK \circ bK = (ab)K$$

forms the group G/K . This operation is well defined in that if $a' \in aK$ and $b' \in bK$, then $a'b' \in abK$, and $a'b'K = abK$. This is also analogously defined for right cosets.

Proof. Let $X, Y \in G/K$ with $Z = XY \in G/K$. Then X, Y, Z are left cosets of K . Since K is the kernel of a homomorphism, then $X = \varphi^{-1}(a)$ and $Y = \varphi^{-1}(b)$ for $a, b \in H$. Moreover, $Z = \varphi^{-1}(ab)$. Let $x \in X$ and $y \in Y$ so that $\varphi(x) = a$ and $\varphi(y) = b$. Then $X = xK$ and $Y = yK$. Then $xy \in Z$ if and only if $xy \in \varphi^{-1}(ab)$, or $\varphi(x)\varphi(y) = ab$, hence $Z = xyK$. Exercise 3.1.2 shows the other direction. Lastly, Proposition 3.2 shows that $xK = Kx$ and $yK = Ky$ for every $x, y \in G$. ■

► **NOTE:** Using similar notation from $\mathbb{Z}/n\mathbb{Z}$, we denote $G/K = \bar{G}$ with $\bar{g} = gK$ and $\bar{g}\bar{h} = \overline{gh}$.

♠ **PROPOSITION 3.4: SET OF COSETS IS PARTITION, EQUIVALENCY CRITERION OF COSETS**

Let $N \leq G$. The set of left cosets of N in G form a partition of G . Moreover, for all $g, h \in G$, then $gN = hN$ if and only if $h^{-1}g \in N$. In particular, $gN = hN$ if and only if g and h represent the same coset.

Proof. It is clear that $g \in gN$ for every $g \in G$. If $gN \cap hN$ is nonempty, let $k \in gN \cap hN$. Then $k = gn_1 = hn_2$ for $n_1, n_2 \in N$. Note that $g = kn_1^{-1} = hn_2n_1^{-1}$ where $n_2n_1^{-1} \in N$ since $N \leq G$. Then for any $gn \in gN$ for some $n \in N$, we have $gn = hn_2n_1^{-1}n \in hN$ so that $gN \subseteq hN$. A similar argument shows that $hN \subseteq gN$, hence they are equal, so that the cosets are either distinct or equal.

The preceding argument shows $gN = hN$ if and only if $g \in hN$, or $g = hn$, or $h^{-1}g \in N$. Moreover, $h \in gN$ implies that h is a representative of gN , so that $hN = gN$ if and only if g and h represent the same coset. ■

♠ **PROPOSITION 3.5: COSET MULTIPLICATION, EXTENDED**

Let G be a group with $N \leq G$.

- (i) The operation $\cdot$ on the set of left cosets of N in G described by

$$aN \cdot bN = (ab)N$$

is well defined if and only if $gng^{-1} \in N$ for every $g \in G$ and $n \in N$.

- (ii) Well definedness of the above operation implies that the set of left cosets of N in G forms a group with identity $1N = N$ and inverse of gN given by $(gN)^{-1} = g^{-1}N$.

Proof.

- (i) Suppose the operation is well defined, and let $g \in G$ and $n \in N$. Then $g^{-1}N = ng^{-1}N$. Since $1 \in N$, then $ng^{-1}1 \in ng^{-1}N$ so that $ng^{-1} = g^{-1}n'$ for some $n' \in N$. Then $gng^{-1} \in N$.

If $gng^{-1} \in N$ instead for every $g \in G$ and $n \in N$, let $a, a_0 \in aN$ and $b, b_0 \in bN$ so that $a_0 = an$ and $b_0 = bm$ for some $n, m \in N$. Then

$$a_0b_0 = (an)(bm) = a(nb)m = a(bn')m$$

where we note that $nb = b(n')$ for some $n' \in N$ by assumption. Then $a_0b_0 \in abN$ so that $a_0b_0N \cap abN$ is nonempty. By Proposition 3.4, then $a_0b_0N = abN$, hence the operation is well defined.

- (ii) Trivial verification using the prior proposition and the fact that G is a group. ■

♦ **DEFINITION:** Let $N \leq G$ for group G .

- For $n \in N$, the **conjugate** of n by g is the element gng^{-1} .
- The **conjugate** of N by g is the set $gNg^{-1} = \{gng^{-1} \mid n \in N\}$.
- We say some $g \in G$ **normalizes** N if $gNg^{-1} = N$.
- We say N is a **normal subgroup** of G , denoted by $N \trianglelefteq G$, if $gNg^{-1} = N$ for every $g \in G$. Equivalently, N is normal if and only if every element of G normalizes N .

♥ **THEOREM 3.6: CHARACTERIZATIONS OF NORMAL SUBGROUPS**

Let G be a group with $N \leq G$. Then the following are equivalent:

- $N \trianglelefteq G$,
- $N_G(N) = G$,
- $gN = Ng$ for every $g \in G$,
- the operation on left cosets of N in G described in Proposition 3.5 makes the set of left cosets into a group,
- $gNg^{-1} \subseteq N$ for every $g \in G$.

Proof.

- (i) if and only if (ii) by definition of normality and normalizer.
- (ii) if and only if (iii), since $gNg^{-1} = N$ implies $gN = Ng$ for every $g \in G$.
- (iii) implies (iv) by Proposition 3.5.
- (iv) implies (v), as well definedness of the operation implies $gng^{-1} \in N$ for every $g \in G$ and $n \in N$ by Proposition 3.5.
- (v) implies (i), because $g^{-1}Ng \subseteq N$ in particular, hence $N \subseteq gNg^{-1} \subseteq N$ so that $gNg^{-1} = N$ for every $g \in G$. ■

♠ **PROPOSITION 3.7: NORMAL SUBGROUP ARE KERNELS**

$N \trianglelefteq G$ if and only if $N = \ker \varphi$ for some homomorphism φ on G .

Proof. If $N \trianglelefteq G$, let $\pi : G \rightarrow G/N$ be the map

$$\pi(g) = gN, \quad g \in G$$

It is clear that π is a homomorphism. Moreover, $\pi(g) = 1N$ when $gN = N$, or $g \in N$, hence $\ker \pi = N$.

If N is instead a kernel of some homomorphism, then $gN = Ng$ for $g \in G$ by [Proposition 3.2](#). Then $N \trianglelefteq G$ by [Theorem 3.6](#). ■

♦ **DEFINITION:** For $N \trianglelefteq G$, then the homomorphism $\pi : G \rightarrow G/N$ given by $\pi(g) = gN$ is called the **natural projection homomorphism** from G to G/N . Moreover, for $\bar{H} \leq G/N$, the set $\pi^{-1}(\bar{H})$ is the **complete preimage** of $\bar{H}$ in G .

► **NOTE:** We claim quotient groups of cyclic groups are cyclic. Let $G = Z_n$ for some $n \in \mathbb{Z}^+$ with $g \in G$ such that $\langle g \rangle = G$. For some $N \leq G$, we know $N = \langle g^d \rangle$ for smallest positive integer d such that $g^d \in N$. Note that $g^\alpha N = (gN)^\alpha$ for every $\alpha \in \mathbb{Z}$ by Exercise 3.1.4, hence $G/N = \langle gN \rangle$. By Exercise 3.1.5, we know $|gN| = d$, and $d = |G|/|N|$ by [Proposition 2.5](#), so that $G/N \cong Z_d$.

3.2 More on Cosets and Lagrange's Theorem

♥ THEOREM 3.8: LAGRANGE'S THEOREM

Let G be a finite group with $H \leq G$. Then $|H|$ divides $|G|$. Also, the number of left cosets of H in G is $|G|/|H|$.

Proof. Let $|H| = n$ and let k be the number of left cosets of H in G . Proposition 3.4 shows that these cosets partition H , and the map $H \rightarrow gH$ given by $h \mapsto gh$ is a bijection for every $g \in G$, hence each coset has n elements. Since the cosets partition G , then $|G| = kn$, so that n divides $|G|$ and the number of left cosets is $k = |G|/n$. ■

♦ **DEFINITION:** Let G be a (possibly infinite) group with $H \leq G$. The **index** of H in G is the number of left cosets of H in G , denoted by $|G : H|$.

♣ COROLLARY 3.9: ELEMENT ORDER DIVIDES GROUP ORDER

Let G be a finite group and let $g \in G$. Then the order of g divides the order of G . In particular, $g^{|G|} = 1$.

Proof. Proposition 2.2 shows that $|g| = |\langle x \rangle|$. Then the first part follows by using Lagrange's Theorem on $H = \langle g \rangle$. Since $|G|$ is a multiple of $|g|$, the second part follows. ■

♣ COROLLARY 3.10: PRIME ORDER IMPLIES CYCLIC

Let G be a group of prime order. Then $G \cong Z_p$ for some prime p .

Proof. For non-identity $g \in G$, then $\langle g \rangle \leq G$ is nontrivial, hence $\langle g \rangle = G$ by the primality of $|G|$. Then G is cyclic, and $G \cong Z_{|G|}$ by Theorem 2.4. ■

► **NOTE:** We claim $H \trianglelefteq G$ when $|G : H| = 2$. For $g \in G - H$, then $G/H = \{1H, gH\} = \{H1, Hg\}$. Since $H = 1H = H1$, then $gH = G - H = Hg$, hence $H \trianglelefteq G$ by Theorem 3.6.

♥ THEOREM 3.11: CAUCHY'S THEOREM

Let G be finite with prime p dividing $|G|$. Then there exists $g \in G$ where $|g| = p$.

♥ THEOREM 3.12: SYLOW

Let G be finite with $|G| = p^n m$ such that p is prime and $p \nmid m$. Then there exists $H \leq G$ such that $|H| = p^n$.

♦ **DEFINITION:** Let $H, K \leq G$. The **subgroup product** of H and K is the set

$$HK = \{hk \mid h \in H, k \in K\}$$

♠ PROPOSITION 3.13: CARDINALITY OF SUBGROUP PRODUCT

Let H and K be subgroups of a finite group G . Then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

Proof. Observe HK is a union of left cosets of K . To find the number of distinct left cosets of the form hK for $h \in H$, observe that $h_1K = h_2K$ for some $h_1, h_2 \in H$ if and only if $h_2^{-1}h_1 \in K$, or $h_2^{-1}h_1 \in H \cap K$. Then the number of distinct left cosets of K in HK is $|H : H \cap K| = |H|/|H \cap K|$. The result follows by noting that each coset has $|K|$ elements. ■

♠ PROPOSITION 3.14: CHARACTERIZATION OF SUBGROUP PRODUCT AS A SUBGROUP

If $H, K \leq G$, then $HK \leq G$ if and only if $HK = KH$.

Proof. If $HK \leq G$, we have $KH \subseteq HK$ from $H, K \leq HK$. For $hk \in HK$, then $hk = (h_0k_0)^{-1} = k_0^{-1}h_0^{-1} \in KH$ for some $h_0 \in H$ and $k_0 \in K$, hence $HK \subseteq KH$.

If $HK = KH$, let $a, b \in HK$ with $a = h_1k_1, b = h_2k_2$ for $h_i \in H$ and $k_i \in K$. Then $ab^{-1} = h_1k_1k_2^{-1}h_2^{-1} \in HK$. Since $KH = HK$, $(k_1k_2^{-1})h_2^{-1} = h_3k_3$ for some $h_3 \in H$ and $k_3 \in K$, hence $ab^{-1} = h_1h_3k_3 \in HK$, or $HK \leq G$. ■

♣ COROLLARY 3.15: NORMALIZING CRITERION FOR SUBGROUP PRODUCTS

Let $H, K \leq G$ with $H \leq N_G(K)$. Then $HK \leq G$. In particular, $K \trianglelefteq G$ implies $HK \leq G$ for every $H \leq G$.

Proof. Observe $khk^{-1} \in K$ for all $h \in H$ and $k \in K$ by assumption. Then $hk = (khk^{-1})h \in KH$, and $kh = h(h^{-1}kh) \in HK$ for all $h \in H$ and $k \in K$, hence $HK = KH$. By Proposition 3.14, then $HK \leq G$. ■

♦ **DEFINITION:** Let $A \subseteq N_G(K)$ or $C_G(K)$. Then A **normalizes**, or **centralizes** K , respectively.

3.3 The Isomorphism Theorems

♥ THEOREM 3.16: THE FIRST ISOMORPHISM THEOREM

If $\varphi : G \rightarrow H$ is a group homomorphism, then $\ker \varphi \trianglelefteq G$ and $G/\ker \varphi \cong \varphi(G)$.

Proof. Note that $\ker \varphi \trianglelefteq G$ by Proposition 3.7. Consider the map

$$\psi : G/\ker \varphi \rightarrow \varphi(G), \quad g \ker \varphi \mapsto \varphi(g),$$

and also observe that $\varphi = \psi \circ \pi$ where π is the natural projection homomorphism. To show that ψ is well defined, observe that $g \ker \varphi = h \ker \varphi$ implies $h^{-1}g \in \ker \varphi$, or $\varphi(h) = \varphi(g)$. The backwards shows injectivity, and surjectivity is clear by definition of $\varphi(G)$. A composition of homomorphisms is a homomorphism, hence ψ is an isomorphism from $G/\ker \varphi$ to $\varphi(G)$. ■

♣ COROLLARY 3.17: CONSEQUENCES OF THE FIRST ISOMORPHISM THEOREM

Let $\varphi : G \rightarrow H$ be a group homomorphism. Then

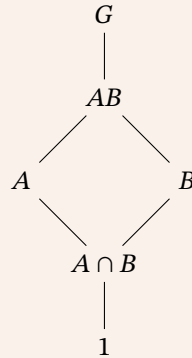
- (i) φ is injective if and only if $\ker \varphi = 1$, and
- (ii) $|G : \ker \varphi| = |\varphi(G)|$.

Proof.

- (i) If φ is injective, then $\varphi(g) = 1$ implies $g = 1$. Conversely, a trivial kernel implies for any $g, h \in G$ such that $\varphi(g) = \varphi(h)$ that $\varphi(h^{-1}g) = 1$, hence $h^{-1}g = 1$ or $g = h$.
- (ii) By the First Isomorphism Theorem, then $G/\ker \varphi \cong \varphi(G)$, so that $|G : \ker \varphi| = |G/\ker \varphi| = |\varphi(G)|$. ■

♥ THEOREM 3.18: THE SECOND (DIAMOND) ISOMORPHISM THEOREM

Let G be a group with $A, B \leq G$ and $A \leq N_G(B)$. Then $AB \leq G$, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$. A diagram of the subgroup lattice is as follows:



Proof. Corollary 3.15 shows that $AB \leq G$. Also, $AB \leq N_G(B)$, or $B \trianglelefteq AB$. Consider $\pi|_A : AB \rightarrow AB/B$. Then $\ker \pi|_A$ consists of elements $a \in A$ such that $aB = B$, or $a \in B$, hence $a \in A \cap B$. By the First Isomorphism Theorem, then $A \cap B \trianglelefteq A$ and $A/A \cap B \cong AB/B$. ■

♥ THEOREM 3.19: THE THIRD ISOMORPHISM THEOREM

Let G be a group with $N_1 \trianglelefteq G$ and $N_2 \trianglelefteq G$ such that $N_1 \leq N_2$. Then $N_2/N_1 \trianglelefteq G/N_1$ and

$$(G/N_1)/(N_2/N_1) \cong G/N_2.$$

Proof. It is clear that $N_2/N_1 \trianglelefteq G/N_1$ by $N_2 \trianglelefteq G$. The map

$$\varphi : G/N_1 \rightarrow G/N_2, \quad gN_1 \mapsto gN_2$$

is well defined, since $N_1 \leq N_2$. It is easy to see that φ is a surjective homomorphism. Moreover, $\varphi(gN_1) = N_2$ if and only if $gN_2 = N_2$, or $g \in N_2$, hence $\ker \varphi = N_2/N_1$. The First Isomorphism Theorem finishes the proof. ■

♥ THEOREM 3.20: THE FOURTH (LATTICE) ISOMORPHISM THEOREM

Let G be a group with $N \trianglelefteq G$. Then every subgroup of $\bar{G}$ is of the form $\bar{A}$ for some $A \leq G$ with $N \leq A$, and $A \mapsto \bar{A}$ is a bijection from the set of subgroups of G containing N to the set of subgroups of $\bar{G}$. This bijection has the following properties:

- (i) $A \leq B$ if and only if $\bar{A} \leq \bar{B}$,
- (ii) $A \leq B$ implies $[B : A] = [\bar{B} : \bar{A}]$,
- (iii) $\langle A, B \rangle = \langle \bar{A}, \bar{B} \rangle$,
- (iv) $A \cap B = \bar{A} \cap \bar{B}$,
- (v) $A \trianglelefteq G$ if and only if $\bar{A} \trianglelefteq \bar{G}$.

Proof. See Exercise 3.3.2. ■

► **NOTE:** A criterion to define homomorphisms on quotient groups is requiring that the kernel of the homomorphism contains the normal subgroup. In particular, if $N \trianglelefteq G$ and $\varphi : G \rightarrow H$ is a homomorphism such that $N \subseteq \ker \varphi$, then we define a homomorphism $\psi : G/N \rightarrow H$ by $\psi(gN) = \varphi(g)$ for $g \in G$. The well definedness of ψ follows from the fact that $gN = hN$ implies $h^{-1}g \in N \subseteq \ker \varphi$, hence $\varphi(g) = \varphi(h)$. The diagram below commutes, since $\psi \circ \pi = \varphi$ where $\pi : G \rightarrow G/N$ is the natural projection homomorphism.

$$\begin{array}{ccc}
 G & \xrightarrow{\pi} & G/N \\
 & \searrow \varphi & \downarrow \psi \\
 & & H
 \end{array}$$

3.4 Composition Series and the Hölder Program

♠ PROPOSITION 3.21: CAUCHY'S THEOREM FOR ABELIAN GROUPS

Let G be finite abelian with prime p dividing $|G|$. Then there exists $g \in G$ where $|g| = p$.

Proof. Induct on $|G|$. Suppose $|G| > 1$ and $g \in G$ with $g \neq 1$. If $|G| = p$, then $|g| = p$ by Lagrange's. If $|G| > p$ and p divides $|g|$, then $|g| = kp$ for $k \in \mathbb{Z}^+$, hence $|g^k| = p$ by Proposition 2.5.

Suppose p does not divide $|g|$, and put $N = \langle g \rangle$. Since G is abelian, $N \trianglelefteq G$, and Lagrange's shows that $|G/N| = |G|/|N|$ with $|G/N| < |G|$. Since p does not divide $|N|$, then p divides $|G/N|$. By the inductive hypothesis, there exists $\bar{h} \in G/N$ with $|\bar{h}| = p$. Note that $h \notin N$ while $h^p \in N$ since $|\bar{h}| = p$, $\langle h^p \rangle \neq \langle h \rangle$, or $|h^p| < |h|$. By Proposition 2.5, we have p divides $|h|$. The proof is finished by the first case. ■

♦ **DEFINITION:** A (finite/infinite) group G is **simple** if $G \neq 1$ and the only normal subgroups of G are 1 and G .

♦ **DEFINITION:** Let G be a group with a sequence of subgroups

$$1 = N_0 \leq N_1 \leq \cdots \leq N_{s-1} \leq N_s = G$$

such that $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is simple for every $0 \leq i < s$. Then the sequence is a **composition series** of G . Moreover, the quotient groups N_{i+1}/N_i are called the **composition factors** of G .

♥ THEOREM 3.22: JORDAN-HÖLDER

Let G be nontrivial and finite. Then

- (i) G has a composition series, and
- (ii) the composition factors in a composition series are unique in that if

$$1 = N_0 \leq N_1 \leq \cdots \leq N_{s-1} \leq N_s = G, \quad 1 = M_0 \leq M_1 \leq \cdots \leq M_{t-1} \leq M_t = G$$

are two composition series of G , then $s = t$ and there exists a permutation π of $\{1, 2, \dots, s\}$ such that $N_{i+1}/N_i \cong M_{\pi(i)+1}/M_{\pi(i)}$ for all i .

Proof. See Exercises 3.4.6, 3.4.9, and 3.4.10. ■

♦ **DEFINITION:** The **Hölder Program** is the program:

- (i) Classify all simple groups, and
- (ii) Use the classification of simple groups to classify all finite groups by the Jordan-Hölder Theorem.

♥ THEOREM: THE CLASSIFICATION OF FINITE SIMPLE GROUPS

There are 18 infinite families of finite simple groups, and 26 sporadic finite simple groups such that every finite simple group is isomorphic to exactly one of these groups.

♥ THEOREM: FEIT-THOMPSON

Every finite simple group of odd order is isomorphic to Z_p for some prime p .

♦ **DEFINITION:** A group G is **solvable** if there is a chain of subgroups

$$1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_{n-1} \trianglelefteq H_n = G$$

such that H_{i+1}/H_i is abelian for every $0 \leq i < n$.

♥ THEOREM: HALL'S THEOREM AND ITS CONVERSE

A finite group G is solvable if and only if for every divisor n of $|G|$ such that $\gcd(n, |G|/n) = 1$, there exists a subgroup $H \leq G$ such that $|H| = n$.

✓ **EXAMPLE:** We claim for any group G with $N \trianglelefteq G$ solvable and G/N solvable, then G is solvable. Let $1 = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_n = N$ be a chain of subgroups such that N_{i+1}/N_i is abelian for every $0 \leq i < n$. Let $\bar{1} = \bar{H}_0 \trianglelefteq \bar{H}_1 \trianglelefteq \cdots \trianglelefteq \bar{H}_m = G/N$ be a chain of subgroups such that $\bar{H}_{i+1}/\bar{H}_i$ is abelian for every $0 \leq i < m$. By the Fourth Isomorphism Theorem, there are subgroups H_i with $N \leq H_i$ such that $\bar{H}_i = H_i/N$. The Third Isomorphism Theorem shows that

$$\bar{H}_{i+1}/\bar{H}_i \cong H_{i+1}/H_i$$

Then

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_n = N = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_m = G$$

is a chain of subgroups with abelian successive quotients, hence G is solvable.

3.5 Transpositions and the Alternating Group

◆ **DEFINITION:** A **transposition** is a 2-cycle. In particular, any $(a_1 a_2 \cdots a_k)$ with $k \geq 2$ can be written as a product of $k - 1$ transpositions:

$$(a_1 a_2 \cdots a_k) = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_2)$$

◆ **DEFINITION:** Let $a_1, a_2, \dots, a_n$ be independent symbols, and consider the **Vandermonde Polynomial** in $a_1, a_2, \dots, a_n$ defined by

$$\Delta = \prod_{1 \leq i < j \leq n} (a_i - a_j)$$

Let $\sigma \in S_n$ act on Δ by permuting the subscripts of a_i . Then $\sigma(\Delta) = \pm 1 \Delta$. Define the **sign** of σ to be

$$\epsilon(\sigma) = \begin{cases} 1 & \text{if } \sigma(\Delta) = +1\Delta \\ -1 & \text{if } \sigma(\Delta) = -1\Delta \end{cases}$$

An **even** permutation is when $\epsilon(\sigma) = 1$, and an **odd** permutation is when $\epsilon(\sigma) = -1$.

♠ **PROPOSITION 3.23: SIGN IS HOMOMORPHISM**

The map $\epsilon : S_n \rightarrow \{\pm 1\}$ is a homomorphism, where $\{\pm 1\} \cong Z_2$.

Proof. For $\tau, \sigma \in S_n$, we have

$$(\tau\sigma)(\Delta) = \prod_{1 \leq i < j \leq n} (a_{\tau\sigma(i)} - a_{\tau\sigma(j)})$$

If $\sigma(\Delta)$ has k factors switched, then $\epsilon(\sigma) = (-1)^k$. There are k factors of the form $a_{\tau\sigma(i)} - a_{\tau\sigma(j)}$ with $\tau\sigma(i) > \tau\sigma(j)$, and τ switches these factors if and only if $\tau(i) > \tau(j)$, so that $\epsilon(\tau\sigma) = (-1)^k \epsilon(\tau) = \epsilon(\sigma)\epsilon(\tau)$. ■

♠ **PROPOSITION 3.24: TRANSPOSITIONS ARE ODD, ϵ IS SURJECTIVE**

Proof. Note that $\epsilon((1\ 2)) = -1$. For any $(i\ j)$ with $1 \leq i < j \leq n$, let $\sigma \in S_n$ such that $\sigma(1) = i$ and $\sigma(2) = j$. Then $\epsilon((i\ j)) = \epsilon(\sigma(1\ 2)\sigma^{-1}) = \epsilon(\sigma)\epsilon((1\ 2))\epsilon(\sigma^{-1}) = -1$. Moreover, ϵ is surjective since $\epsilon((1\ 2)) = -1$ and $\epsilon(\text{id}) = 1$. ■

► **NOTE:** A useful conjugation formula is $\sigma(i\ j)\sigma^{-1} = (\sigma(i)\ \sigma(j))$ for $\sigma \in S_n$ and $1 \leq i < j \leq n$.

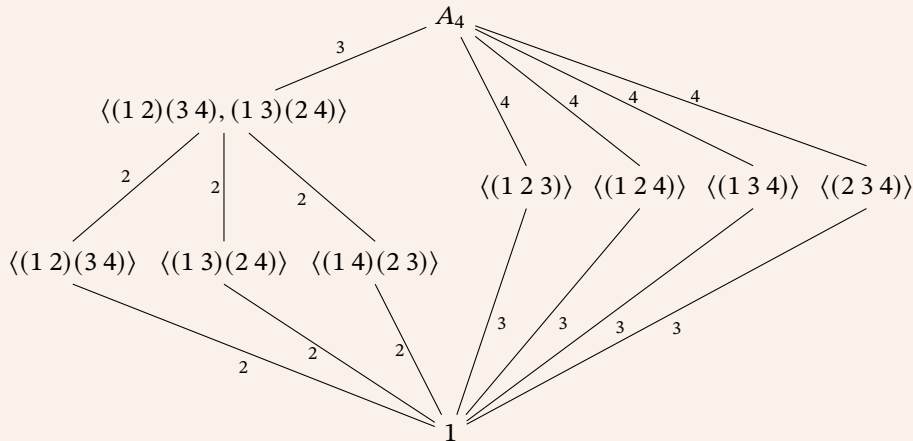
◆ **DEFINITION:** The **alternating group** A_n is the set of even permutations in S_n , or $\ker \epsilon$. By the First Isomorphism Theorem, then $A_n \leq S_n$ and $S_n/A_n \cong Z_2$, hence $|A_n| = n!/2$.

♠ **PROPOSITION 3.25: ODD PERMUTATION IF AND ONLY IF ODD NUMBER OF EVEN LENGTH CYCLES**

A permutation σ is odd if and only if the number of cycles of even length in its disjoint cycle decomposition is odd.

Proof. For $\sigma \in S_n$ with cycle decomposition $\sigma_1\sigma_2 \cdots \sigma_k$, then $\epsilon(\sigma) = \epsilon(\sigma_1)\epsilon(\sigma_2) \cdots \epsilon(\sigma_k)$. For $\epsilon(\sigma) = -1$, there must be an odd number of σ_i with $\epsilon(\sigma_i) = -1$, or an odd number of cycles of even length in the cycle decomposition. Conversely, an odd number of cycles of even length in the cycle decomposition implies $\epsilon(\sigma) = -1$. ■

✓ **EXAMPLE:** We claim the subgroup lattice of A_4 is as follows:



4 Group Actions

4.1 Group Actions and Permutation Representations

► **NOTE:** Please recall definitions of group actions, permutation representations, kernels, stabilizers, and faithful actions from [Section 1.7](#) and [Section 2.2](#).

✓ **EXAMPLE:** Let D_8 act on $A = \{\{1, 3\}, \{2, 4\}\}$. Let $\mathbf{1} = \{1, 3\}$ and $\mathbf{2} = \{2, 4\}$. A rotation sends one pair to the other, while a reflection fixes both pairs. Hence, the permutation representation afforded by this action is

$$r \mapsto (\mathbf{1} \ \mathbf{2}), \quad s \mapsto \text{id},$$

or $\sigma_r = (\mathbf{1} \ \mathbf{2})$ and $\sigma_s = \text{id}$. Note that $r^2 \mapsto \text{id}$, hence the kernel of the action is $\langle r^2, s \rangle$.

♠ **PROPOSITION 4.1: GROUP ACTIONS AS HOMOMORPHISMS**

For a group G acting on a nonempty set A , there exists a bijection between the set of group actions of G on A and the set of homomorphisms from G to S_A .

Proof. Let $\varphi : G \rightarrow S_A$, and consider the action of G on A defined by $g \cdot a = \varphi(g)(a)$ for all $g \in G$ and $a \in A$. Clearly, the kernel of this action is $\ker \varphi$, and the permutation representation afforded by this action is φ itself. ■

♦ **DEFINITION:** Let G be a group. Then a **permutation representation** of G is a homomorphism of G into S_A for nonempty A . A given action of G on A **affords** or **induces** the associated permutation representation.

♠ **PROPOSITION 4.2: ORBIT-STABILIZER THEOREM**

Let G be a group acting on the nonempty set A . Then the relation on A defined by

$$a \sim b \quad \text{if and only if} \quad a = g \cdot b \text{ for some } g \in G$$

is an equivalence relation. Moreover, for each $a \in A$, the size of the orbit of a is given by $|G : G_a|$, the index of the stabilizer of a in G .

Proof. Let $a, b, c \in G$. Reflexivity is trivial. If $a \sim b$, or $a = g \cdot b$ for $g \in G$, then $g^{-1} \cdot a = 1 \cdot b$, hence $b \sim a$. If $a \sim b$ and $b \sim c$ with $a = g \cdot b$ and $b = h \cdot c$ for $g, h \in G$, then $a = (gh) \cdot c$. Hence, $\sim$ is an equivalence relation.

Consider $[a]$. Let $b = g \cdot a \in [a]$. Then $g[a]$ is a left coset of G_a in G . Define the map

$$\psi : [a] \rightarrow G/G_a, \quad g \cdot a \mapsto gG_a.$$

To see that ψ is well defined and injective, note that $g \cdot a = h \cdot a$ if and only if $gG_a = hG_a$. Surjectivity is clear. ■

♦ **DEFINITION:** Let G act on nonempty A .

(i) The **orbit** of $a \in A$ is the equivalence class $[a]$ in [Proposition 4.2](#).

(ii) The action is **transitive** if there is only one orbit, i.e., for every $a, b \in A$, there exists some $g \in G$ such that $g \cdot a = b$.

✓ **EXAMPLE:** S_n acts on $\{1, 2, \dots, n\}$ transitively by the natural action.

► **NOTE:** To see that elements of S_n have unique cycle decompositions, let $A = \{1, 2, \dots, n\}$, $\sigma \in S_n$, and $G = \langle \sigma \rangle$. Let G act on A , so by [Proposition 4.2](#), A is partitioned into a unique set of disjoint orbits. Let O be an orbit with some $a \in O$. The proof of [Proposition 4.2](#) shows that the cosets of G_a in G are in bijection with elements of O explicitly by

$$\psi : O \rightarrow G/G_a, \quad \sigma^k \cdot a \mapsto \sigma^k G_a.$$

Note that G/G_a is cyclic with order $d = |G : G_a| = |O|$, where d is smallest positive integer such that $\sigma^d \in G_a$. Thus, the distinct cosets of G_a in G are

$$\{G_a, \sigma G_a, \sigma^2 G_a, \dots, \sigma^{d-1} G_a\},$$

and the elements of O are

$$\{a, \sigma \cdot a, \sigma^2 \cdot a, \dots, \sigma^{d-1} \cdot a\}.$$

Then σ acts on O as a d -cycle, and the cycle decomposition of σ is obtained by considering all distinct orbits of G on A . Since the orbits are unique, the cycle decomposition of σ is unique up to the order of the cycles.

4.2 Groups Acting on Themselves by Left Multiplication—Cayley's Theorem

► **NOTE:** For finite G , enumerate the elements as $g_1, \dots, g_n$. A permutation σ_g for $g \in G$ is then given by

$$\sigma_g(i) = j \quad \text{if and only if} \quad gg_i = g_j,$$

with different labelling of the elements of G giving different permutations.

✓ **EXAMPLE:** Label $V_4 = \{1, a, b, c\}$ as 1, 2, 3, 4 respectively. Computing σ_a , we see

$$\begin{aligned} a \cdot 1 &= a \implies \sigma_a(1) = 2, \\ a \cdot a &= 1 \implies \sigma_a(2) = 1, \\ a \cdot b &= c \implies \sigma_a(3) = 4, \\ a \cdot c &= b \implies \sigma_a(4) = 3, \end{aligned}$$

hence $\sigma_a = (1\ 2)(3\ 4)$. Similarly, we have $\sigma_b = (1\ 3)(2\ 4)$ and $\sigma_c = (1\ 4)(2\ 3)$.

✓ **EXAMPLE:** Let $H \leq G$ for group G , and let $A = G/H$. We may define G to act on A by left multiplication via the action

$$g \cdot (xH) = (gx)H \quad \text{for all } g, x \in G.$$

Moreover, if $|G : H|$ is finite, we may similarly enumerate them as previously done.

♥ **THEOREM 4.3: PROPERTIES OF THE COSET ACTION**

Let G be a group with $H \leq G$, and let G act on $A = G/H$ by left multiplication. Let π_H be the associated permutation representation afforded by this action. Then

- (i) G acts transitively on A ,
- (ii) $G_{1H} = H$, and
- (iii) $\ker \pi_H = \bigcap_{g \in G} gHg^{-1}$, and $\ker \pi_H$ is the largest normal subgroup of G contained in H .

Proof.

- (i) Let $g_1H, g_2H \in A$, and put $g = g_2g_1^{-1}$. Then $g \cdot (g_1H) = g_2H$.
- (ii) By definition, $G_{1H} = \{g \in G \mid g \cdot 1H = 1H\} = \{g \in G \mid gH = H\} = H$.
- (iii) By definition, $g \in \ker \pi_H$ when $gxH = xH$, or $x^{-1}gxH = H$, hence $g \in xHx^{-1}$ for all $x \in G$. Clearly, $\ker \pi_H \trianglelefteq G$ and $\ker \pi_H \leq H$. Let $N \trianglelefteq G$ such that $N \leq H$. Then $N = xNx^{-1} \leq xHx^{-1}$ for all $x \in G$. It follows that $\ker \pi_H$ is the largest normal subgroup of G contained in H . ■

♣ **COROLLARY 4.4: CAYLEY'S THEOREM**

Every group is isomorphic to a subgroup of a symmetric group.

Proof. Use $H = 1$ in [Theorem 4.3](#). Then $\ker \pi_H = 1$, so that π_H is injective, and G is isomorphic to the subgroup $\pi_H(G)$ of $S_{G/H} = S_G$. ■

♣ **COROLLARY 4.5: SUBGROUPS OF SMALLEST PRIME INDEX ARE NORMAL**

Let $|G| = n < \infty$, and let p be the smallest prime dividing n . Then every subgroup of index p in G is normal.

Proof. Let $H \leq G$ with $|G : H| = p$. Let π_H be the permutation afforded by G acting on G/H by left multiplication, let $K = \ker \pi_H$, and put $|H : K| = k$. Then $|G : K| = |G : H||H : K| = pk$. By the First Isomorphism Theorem, $G/K \cong \pi_H(G) \leq S_{G/H} = S_p$. Lagrange's shows that $pk \mid p!$, or $k \mid (p-1)!$. Since p is the smallest prime dividing n , then $k = 1$, so that $H = K \trianglelefteq G$. ■

4.3 Groups Acting on Themselves by Conjugation—The Class Equation

◆ **DEFINITION:**

- Let $g, h \in G$. Then g and h are **conjugate** in G if there exists some $x \in G$ such that $h = xgx^{-1}$. The equivalence classes of this relation are the **conjugacy classes** of G .
- Let G act on $\mathcal{P}(G)$. For $S, T \in \mathcal{P}(G)$, we say S and T are **conjugate** in G if there exists some $x \in G$ such that $T = xSx^{-1}$, generalizing the previous definition of conjugacy for elements of G .

♠ **PROPOSITION 4.6: CONJUGATION UNDER ORBIT-STABILIZER THEOREM**

The number of conjugates of $S \subseteq G$ in G is $|G : N_G(S)|$. Moreover, the number of conjugates of $g \in G$ is $|G : C_G(g)|$.

Proof. Note that $G_S = \{g \in G \mid gSg^{-1} = S\} = N_G(S)$, and $N_G(g) = C_G(g)$ for any $g \in G$. ■

♥ **THEOREM 4.7: THE CLASS EQUATION**

Let G be finite with $g_1, g_2, \dots, g_s$ be representatives of distinct conjugacy classes of G not in $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^s |G : C_G(g_i)|.$$

Proof. Note that each $z \in Z(G)$ has conjugacy class $\{z\}$, and each g_i has conjugacy class of size $|G : C_G(g_i)|$. Since the conjugacy classes of G partition G , the class equation follows. ■

✓ **EXAMPLE:** Observe $\langle g \rangle \leq C_G(g)$ for any $g \in G$. In $G = Q_8$, we know $\langle i \rangle \leq C_G(i) \leq G$. However, $i \notin Z(G)$, so that $C_G(i)$ is a proper subgroup of G . Since $|G : \langle i \rangle| = 4$, it follows that $C_G(i) = \langle i \rangle$. Hence, the conjugacy class of i has size $|G : C_G(i)| = 2$, and the conjugates of i are itself and $-i = jij^{-1}$. The conjugacy classes for j and k are similar, hence the conjugacy classes of Q_8 are

$$\{1\}, \quad \{-1\}, \quad \{\pm i\}, \quad \{\pm j\}, \quad \{\pm k\}$$

with sizes 1, 1, 2, 2, and 2, respectively. The class equation for Q_8 is then $|Q_8| = 2 + 2 + 2 + 2 = 8$.

♥ **THEOREM 4.8: PRIME POWER ORDER IMPLIES NONTRIVIAL CENTER**

Let p be prime with $|P| = p^n$ for $n \geq 1$. Then $Z(P)$ is nontrivial.

Proof. The class equation gives

$$|P| = |Z(P)| + \sum_{i=1}^m |P : C_P(g_i)|,$$

with representatives $g_1, g_2, \dots, g_m$ of distinct conjugacy classes. Since $C_P(g_i) \neq P$, each $|P : C_P(g_i)|$ is a power of p , hence $|Z(P)|$ is divisible by p , so that $Z(P)$ is nontrivial. ■

♣ **COROLLARY 4.9: CLASSIFICATION OF GROUPS OF ORDER p^2**

If $|P| = p^2$ for prime p , then P is abelian. More precisely, $P \cong Z_{p^2}$ or $P \cong Z_p \times Z_p$.

Proof. Theorem 4.8 implies $Z(P) \neq 1$, hence $P/Z(P)$ is cyclic and P is abelian. If $x \in P$ with $|x| = p^2$, then $P = \langle x \rangle$, so assume no element of P has order p^2 , hence has order p . Consider nonidentity $a \in P$ with $b \in P - \langle a \rangle$. We have $\langle a \rangle < \langle a, b \rangle$ strictly by construction, hence $\langle a, b \rangle = P$. Observe $\langle a \rangle \times \langle b \rangle = Z_p \times Z_p$, and the map $(a^m, b^n) \mapsto a^m b^n$ is clearly an isomorphism from $\langle a \rangle \times \langle b \rangle$ to P . ■

♠ **PROPOSITION 4.10: CONJUGATION IN S_n**

Let $\sigma, \tau \in S_n$. If σ has cycle decomposition

$$\sigma = (a_1 a_2 \cdots a_{\ell_1})(b_1 b_2 \cdots b_{\ell_2}) \cdots$$

then the cycle decomposition of $\tau\sigma\tau^{-1}$ is

$$\tau\sigma\tau^{-1} = (\tau(a_1) \tau(a_2) \cdots \tau(a_{\ell_1}))(\tau(b_1) \tau(b_2) \cdots \tau(b_{\ell_2})) \cdots$$

Proof. If $\sigma(i) = j$, it is clear that $\tau\sigma\tau^{-1}(\tau(i)) = \tau(j)$. ■

◆ **DEFINITION:** Let $\sigma \in S_n$ be the product of disjoint cycles with lengths $\ell_1, \ell_2, \dots, \ell_k$, such that $\ell_1 \leq \ell_2 \leq \dots \leq \ell_k$. Then the **cycle type** of σ is the k -tuple $(\ell_1, \ell_2, \dots, \ell_k)$.

◆ **DEFINITION:** A **partition** of $n \in \mathbb{Z}^+$ is a nondecreasing sequence of positive integers with sum n .

♠ **PROPOSITION 4.11: CONJUGACY CRITERION IN S_n**

$\sigma, \tau \in S_n$ are conjugate if and only if they have the same cycle type. Moreover, the number of conjugacy classes of S_n is the number of partitions of n .

Proof. Proposition 4.10 shows that conjugate permutations have the same cycle type. Conversely, if σ and τ have the same cycle type, then we may write

$$\sigma = (a_1 a_2 \cdots a_{\ell_1})(b_1 b_2 \cdots b_{\ell_2}) \cdots, \quad \tau = (c_1 c_2 \cdots c_{\ell_1})(d_1 d_2 \cdots d_{\ell_2}) \cdots$$

Then $\alpha \in S_n$ defined by $\alpha(a_i) = c_i$ for $1 \leq i \leq \ell_1$, $\alpha(b_i) = d_i$ for $1 \leq i \leq \ell_2$, and so on, satisfies $\alpha\sigma\alpha^{-1} = \tau$.

The conjugacy classes of S_n are in bijection with the cycle types of elements of S_n , which are in bijection with the partitions of n , hence the second assertion follows. ■

✓ **EXAMPLE:** For an m -cycle $\sigma \in S_n$, we claim the number of conjugates of σ is

$$\frac{|S_n|}{|C_{S_n}(\sigma)|} = \frac{n!}{m(n-m)!} = \frac{n(n-1) \cdots (n-m+1)}{m}$$

Note σ commutes with its powers, and commutes with the $(n-m)!$ permutations of the remaining $n-m$ integers, hence $|C_{S_n}(\sigma)| = m(n-m)!$.

► **NOTE:** If $H \trianglelefteq G$, then for any conjugacy class C of G , either $C \subseteq H$ or $C \cap H = \emptyset$: if $g \in C \cap H$, then $xgx^{-1} \in H$ for all $x \in G$, hence $C \subseteq H$.

♥ **THEOREM 4.12: A_5 IS SIMPLE**

Proof. We know the representatives of the even permutations are of the form

$$\text{id}, (1\ 2\ 3), (1\ 2\ 3\ 4\ 5), (1\ 2)(3\ 4).$$

- id is its own conjugacy class with size 1.
- $C_{A_5}((1\ 2\ 3)) = \langle (1\ 2\ 3) \rangle$ has order 3, so the conjugacy class of $(1\ 2\ 3)$ has size $60/3 = 20$. All 3-cycles are conjugate in A_5 since they have the same cycle type.
- $C_{A_5}((1\ 2\ 3\ 4\ 5)) = \langle (1\ 2\ 3\ 4\ 5) \rangle$ has order 5, so the conjugacy class of $(1\ 2\ 3\ 4\ 5)$ has size $60/5 = 12$. However, there are 24 5-cycles in A_5 , so there are two conjugacy classes of 5-cycles in A_5 .
- All odd-order elements have been accounted for, hence the remaining 15 are order 2 with cycle type $(2, 2)$. It is easy to verify that $C_{A_5}((1\ 2)(3\ 4))$ has order 4, so the conjugacy class of $(1\ 2)(3\ 4)$ has size $60/4 = 15$.

The class equation for A_5 is then

$$60 = 1 + 20 + 12 + 12 + 15.$$

By the previous note, a nontrivial normal subgroup of A_5 must be the union of some of the nontrivial conjugacy classes, but no such union has size dividing 60, hence A_5 is simple. ■

◆ **DEFINITION:** A **right group action** can be defined analogously as a map from $A \times G$ to A , denoted by $a \cdot g$ for $a \in A$ and $g \in G$, such that

- $a \cdot 1 = a$ for all $a \in A$, and
- $(a \cdot g) \cdot h = a \cdot (gh)$ for all $a \in A$ and $g, h \in G$.

► **NOTE:** Conjugation is typically written as a right group action: $a^g = g^{-1}ag$ for $a, g \in G$. Similarly, $S^g = g^{-1}Sg$ for $S \subseteq G$ and $g \in G$.

► **NOTE:** If we are given a left group action of G on A , we can define a right group action of G on A by $a \cdot g = g^{-1} \cdot a$ for all $a \in A$ and $g \in G$.

4.4 Automorphisms

◆ **DEFINITION:** An isomorphism from a group G to itself is an **automorphism**. The set of automorphisms of G is denoted by $\text{Aut}(G)$.

♠ **PROPOSITION 4.13: CONJUGATION ON NORMAL SUBGROUPS**

Let $H \trianglelefteq G$. Let G act on H by conjugation. Then G acts on conjugation on H by automorphisms, defined as

$$g \cdot h \mapsto ghg^{-1}, \quad g \in G, h \in H.$$

Each conjugation by g is in $\text{Aut}(H)$, and $\pi_H : G \rightarrow \text{Aut}(H)$ is a homomorphism with kernel $C_G(H)$. In particular, $G/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$.

Proof. Let $\varphi_g : H \rightarrow H$ be defined by $\varphi_g(h) = ghg^{-1}$ for all $h \in H$. Since $H \trianglelefteq G$, then φ_g is well defined. Observe φ_g has inverse $(\varphi_g)^{-1} = \varphi_{g^{-1}}$, so φ_g is bijective. To see that φ_g is a homomorphism, note that

$$\varphi_g(h_1h_2) = g(h_1h_2)g^{-1} = (gh_1g^{-1})(gh_2g^{-1}) = \varphi_g(h_1)\varphi_g(h_2)$$

for all $h_1, h_2 \in H$. Hence, $\varphi_g \in \text{Aut}(H)$ for all $g \in G$.

Since π_H has been shown to be a homomorphism, we simply observe that $\ker \pi_H$ consists of $g \in G$ such that $ghg^{-1} = h$ for all $h \in H$, or $g \in C_G(H)$, hence $\ker \pi_H = C_G(H)$. The First Isomorphism Theorem then gives $G/C_G(H) \cong \pi_H(G) \leq \text{Aut}(H)$. ■

♣ **COROLLARY 4.14: CONJUGATE SUBGROUPS ARE ISOMORPHIC**

For $H \leq G$ with $g \in G$, then $H \cong gHg^{-1}$. In particular, $|h| = |ghg^{-1}|$, and $|H| = |gHg^{-1}|$.

Proof. Put $G = H$ in Proposition 4.13. Conjugation is then an automorphism, and the result follows. ■

♣ **COROLLARY 4.15: NORMALIZER/CENTRALIZER THEOREM**

For $H \leq G$, then $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$. In particular, $G/Z(G)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Proof. Put $G = N_G(H)$ in Proposition 4.13. Then $N_G(H)/C_G(H) \cong \pi_H(N_G(H)) \leq \text{Aut}(H)$. The second assertion follows by putting $H = G$, hence $N_G(H) = G$ and $C_G(H) = Z(G)$. ■

◆ **DEFINITION:** Conjugation by $g \in G$ is called an **inner automorphism** of G . The set of inner automorphisms of G is denoted by $\text{Inn}(G)$, which is a subgroup of $\text{Aut}(G)$.

► **NOTE:** Observe $\text{Inn}(G) \cong G/Z(G)$ by Corollary 4.15.

◆ **DEFINITION:** Some $H \leq G$ is called **characteristic**, denoted $H \text{ char } G$, if $\varphi(H) = H$ for all $\varphi \in \text{Aut}(G)$.

► **NOTE:**

- (i) Characteristic subgroups are normal,
- (ii) if H has unique order in G , then $H \text{ char } G$ since automorphisms preserve order, and
- (iii) If $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.

♠ **PROPOSITION 4.16: $\text{Aut}(Z_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$**

For $n \in \mathbb{Z}^+$, then $\text{Aut}(Z_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, where $|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n)$ is the Euler φ -function of n .

Proof. Put $Z_n = \langle x \rangle$. For $\varphi \in \text{Aut}(Z_n)$, then $\varphi(x) = x^k$ for $k \in \mathbb{Z}$, and k determines φ . Denote this map as φ_k . Since φ_k is an automorphism, then x^k is a generator of Z_n , so that $\gcd(k, n) = 1$. Conversely, if $\gcd(k, n) = 1$, then φ_k is an automorphism. We then have a surjection

$$\Phi : \text{Aut}(Z_n) \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times, \quad \varphi_k \mapsto k \pmod{n}.$$

Observe $\varphi_k \circ \varphi_\ell(x) = \varphi_k(x^\ell) = x^{k\ell} = \varphi_{k\ell}(x)$, hence

$$\Phi(\varphi_k \circ \varphi_\ell) = \Phi(\varphi_{k\ell}) = k\ell \pmod{n} = (k \pmod{n})(\ell \pmod{n}) = \Phi(\varphi_k)\Phi(\varphi_\ell),$$

so Φ is a homomorphism. Finally, $\ker \Phi$ is trivial, so Φ is an isomorphism. ■

✓ **EXAMPLE:** If $|G| = pq$ for distinct primes $p \leq q$ such that $p \nmid q - 1$, then G is abelian. If $Z(G) \neq 1$, then $G/Z(G)$ is cyclic, hence G abelian. If $Z(G) = 1$ with every nonidentity element of G having order p , then its centralizer has q , hence the class equation is $pq = 1 + kq$, but $q \nmid 1$. Then there must be some $g \in G$ with $|g| = q$.

Put $H = \langle g \rangle$. Then H has index p in G with p smallest prime dividing $|G|$, so $H \trianglelefteq G$ by [Corollary 4.5](#). Trivial center implies $C_G(H) = H$, hence $G/H = N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, which has order $q - 1$. But [Proposition 4.16](#) has $\text{Aut}(H) = \varphi(q) = q - 1$, so $p \mid q - 1$, a contradiction. Hence, G is abelian.

♠ **PROPOSITION 4.17: COMMON AUTOMORPHISM GROUPS**

- (i) Let p be odd prime with $n \in \mathbb{Z}^+$. Then $\text{Aut}(Z_p)$ is cyclic with order $p - 1$. More generally, $\text{Aut}(Z_{p^n})$ is cyclic with order $p^{n-1}(p - 1)$.
- (ii) For $n \geq 3$, $\text{Aut}(Z_{2^n}) \cong Z_2 \times Z_{2^{n-2}}$ and has a cyclic subgroup of index 2.
- (iii) Let p be prime with abelian additive group V such that $pv = 0$ for all $v \in V$. If $|V| = p^n$, then V is an n -dimensional vector space over $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. More precisely, $\text{Aut}(V)$ is the set of nonsingular linear transformations from V to itself:

$$\text{Aut}(V) \cong \text{GL}(V) \cong \text{GL}_n(\mathbb{F}_p)$$

with $|\text{Aut}(V)| = |\text{GL}_n(\mathbb{F}_p)| = (p^n - 1)(p^n - p) \cdots (p^n - p^{n-1})$.

- (iv) For $n \neq 6$, $\text{Aut}(S_n) = \text{Inn}(S_n) \cong S_n$. However, $n = 6$ is exceptional with $|\text{Aut}(S_6) : \text{Inn}(S_6)| = 2$.
- (v) $\text{Aut}(D_8) \cong D_8$ and $\text{Aut}(Q_8) \cong S_4$.

Proof.

- (i) Check ??.
- (ii) Check ??.
- (iii) Check ?? and ??.
- (iv) Check Exercises 4.4.19 and 6.3.10.
- (v) Check Exercises 4.4.4, 4.4.5, and 6.3.9. ■

✓ **EXAMPLE:** [Chapter 5](#) will define the **elementary abelian group** of order p^n . For a prime p , the elementary abelian group of order p^2 is $Z_p \times Z_p$ with automorphism group $\text{Aut}(Z_p \times Z_p) \cong \text{GL}_2(\mathbb{F}_p)$. Hence, $\text{Aut}(Z_p \times Z_p)$ has order $p(p - 1)^2(p + 1)$. If cyclic, then $\text{Aut}(Z_{p^2})$ has order $p(p - 1)$, so that $\text{Aut}(Z_p \times Z_p)$ is much larger than $\text{Aut}(Z_{p^2})$.

✓ **EXAMPLE:** A group of order $45 = 3^2 \cdot 5$ with normal P of order 3^2 is necessarily abelian: by [Corollary 4.15](#), $G/C_G(P)$ is isomorphic to a subgroup of $\text{Aut}(P)$, which has order 6 or 48, depending on cyclicity of P . Since $|P| = 3^2$, then P is abelian so $P \leq C_G(P)$. Since $|G/C_G(P)|$ divides $|\text{Aut}(P)|$, then $|G/C_G(P)|$ divides 6, so that $C_G(P) = G$, hence $P \leq Z(G)$, so that G is abelian.

4.5 Sylow's Theorem

◆ **DEFINITION:** Let G be a group with prime p .

- A **p -group** is a group of order p^n for $n \in \mathbb{Z}^+$. Subgroups of p -groups are called **p -subgroups**.
- If $|G| = p^n m$ where $p \nmid m$, then a **Sylow p -subgroup** of G is a subgroup of order p^n .
- We denote the set of Sylow p -subgroups of G by $\text{Syl}_p(G)$, and its size by $n_p(G) = |\text{Syl}_p(G)|$.

♥ **THEOREM 4.18: SYLOW'S THEOREM(S)**

Let $|G| = p^n m$ for prime p with $p \nmid m$. Then

- $\text{Syl}_p(G) \neq \emptyset$,
- If $P \in \text{Syl}_p(G)$ and Q is a p -subgroup of G , then $Q \leq gPg^{-1}$ for some $g \in G$. In particular, any two Sylow p -subgroups of G are conjugate.
- $n_p(G) \equiv 1 \pmod{p}$, and $n_p = |G : N_G(P)|$ so that $n_p \mid m$.

★ **LEMMA 4.19: INTERSECTIONS WITH NORMALIZER OF SYLOW p -SUBGROUP**

Let $P \in \text{Syl}_p(G)$ and Q be a p -subgroup of G . Then $Q \cap N_G(P) = Q \cap P$.

Proof. Put $R = N_G(P) \cap Q$. Observe $P \leq N_G(P)$, hence $P \cap Q \leq R$. The reverse inclusion is done by considering PR , which is a subgroup of G since $H \trianglelefteq P$. It has order

$$\frac{|P||R|}{|P \cap R|}$$

by [Corollary 3.15](#). Moreover, observe that R is a p -group since $R \leq Q$, so that $|PR|$ is a power of p . Since both P and $P \cap R$ are p -groups, then $|PR|$ is a p -power. Since $P \leq PR$ and $P \in \text{Syl}_p(G)$, then $PR = P$, so that $R \leq P$. ■

Proof of Sylow's Theorem(s).

- Assume $\text{Syl}_p(G) \neq \emptyset$ for all groups less than $|G|$. If p divides $|Z(G)|$, then $z \in Z(G)$ with $|z| = p$ exists. Put $\bar{G} = G/\langle z \rangle$, hence $|\bar{G}| = p^{n-1}m$. Then $\bar{P} \leq \bar{G}$ with $|\bar{P}| = p^{n-1}$ exists by the inductive hypothesis. Let P be the preimage of $\bar{P}$ under the natural projection $G \rightarrow \bar{G}$. Then P is a subgroup of G with $|P| = p^n$, so $P \in \text{Syl}_p(G)$.

Suppose p doesn't divide $|Z(G)|$. Let $g_1, g_2, \dots, g_k$ be representatives of distinct conjugacy classes of G not in $Z(G)$. Then the class equation gives

$$|G| = |Z(G)| + \sum_{i=1}^k |G : C_G(g_i)|.$$

If p divides all $|G : C_G(g_i)|$, then p would divide $|Z(G)|$, hence assume there exists i such that p doesn't divide $|G : C_G(g_i)|$, and put $C = C_G(g_i)$. Then $|C| = p^n m'$, where $p \nmid m'$. By the inductive hypothesis, C has a Sylow p -subgroup P , which is also a Sylow p -subgroup of G since $|P| = p^n$.

- By (i), $\text{Syl}_p(G) \neq \emptyset$, so let $P \in \text{Syl}_p(G)$. Define $C = \{P_1, P_2, \dots, P_k\}$ as the set of all conjugates of P in G . Let Q be a p -subgroup of G acting on C by conjugation, hence

$$C = O_1 \cup O_2 \cup \dots \cup O_\ell$$

where each O_i is a distinct orbit of Q on C . Relabel elements of C such that $P_i \in O_i$ for $1 \leq i \leq \ell$. By [Proposition 4.6](#), we know $|O_i| = |Q : N_Q(P_i)|$ for $1 \leq i \leq \ell$. Observe $N_Q(P_i) = N_G(P_i) \cap Q = P_i \cap Q$ by [Corollary 4.19](#), hence $|O_i| = |Q : P_i \cap Q|$.

Q was arbitrary, so put $Q = P_1$ so that $|O_1| = 1$. It follows that $P_1 \neq P_i$ for $2 \leq i \leq \ell$, hence $P_1 \cap P_i < P_1$, and $|O_i| = |P_1 : P_1 \cap P_i|$ is a power of p for $2 \leq i \leq \ell$. Since $|C| = k$ is the number of conjugates of P , then

$$k = |O_1| + |O_2| + \dots + |O_\ell| \equiv 1 \pmod{p}.$$

Now, if Q is again any p -subgroup of G not contained in any P_i , then $Q \cap P_i < Q$, so each O_i has order divisible by p , contradicting that $k \equiv 1 \pmod{p}$. Hence, $Q \leq P_i$ for some i , so $Q \leq gPg^{-1}$ for some $g \in G$. If $Q \in \text{Syl}_p(G)$, then Q has order p^n , so $Q = gPg^{-1}$.

- By proof in (ii), we know $C = \text{Syl}_p(G)$, so $n_p = |C| \equiv 1 \pmod{p}$. Moreover, we have that [Proposition 4.6](#) gives $n_p = |G : N_G(P)|$, so $n_p \mid m$ since $|G| = p^n m$. ■

► **NOTE:** Sylow's Theorem (ii) and [Corollary 4.14](#) imply that all Sylow p -subgroups of G are isomorphic.

♣ **COROLLARY 4.20: CHARACTERIZATIONS OF NORMAL SYLOW p -SUBGROUPS**

Let $P \in \text{Syl}_p(G)$. Then the following are equivalent:

- (i) $n_p(G) = 1$,
- (ii) $P \trianglelefteq G$,
- (iii) $P \text{ char } G$,
- (iv) if $S \subseteq G$ such that $|s|$ is a power of p for all $s \in S$, then $\langle S \rangle$ is a p -group.

Proof.

- If (i) is true, observe $gPg^{-1} = P$ for all $g \in G$, so $P \trianglelefteq G$. Conversely, if (ii) is true, then $gPg^{-1} = P$ for all $g \in G$, so $n_p(G) = 1$.
- If (ii) is true, P is the unique Sylow p -subgroup of G , and the result follows by Exercise 4.4.7. Conversely, if (iii) is true, the result follows by Exercise 4.4.6.
- If (i) is true, let $S \subseteq G$ with every $s \in S$ having p -power order. Then $s \in gPg^{-1} = P$ for $g \in G$ by [Sylow's Theorem](#) (ii), hence $\langle S \rangle \leq P$ is a p -group. Conversely, if (iv) is true, put S as the union of all Sylow p -subgroups of G . For $P \in \text{Syl}_p(G)$, then $P \leq \langle S \rangle$. However, P has maximal order, so $\langle S \rangle = P$, hence P is the unique Sylow p -subgroup of G . ■

♦ **DEFINITION:** The p -**primary component** of a finite abelian group G is the Sylow p -subgroup of G containing all elements of G with p -power order.

✓ **EXAMPLE:** Consider groups of order pq with primes $p < q$. Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. We claim $Q \trianglelefteq G$, and if $P \trianglelefteq G$, then $G \cong Z_{pq}$.

- Note $n_q \equiv 1 \pmod q$ and $n_q \mid p$, so $n_q = 1$, hence $Q \trianglelefteq G$.
- Observe $n_p \equiv 1 \pmod p$ and $n_p \mid q$, so $n_p = 1$ or q . If $n_p = 1$, then $P \trianglelefteq G$. In particular, if $p \nmid q - 1$, then $n_p \neq q$ so that $P \trianglelefteq G$.

Now put $P = \langle g \rangle$ and $Q = \langle h \rangle$. Note $G/C_G(P)$ is isomorphic to a subgroup of $\text{Aut}(P)$ with order $p - 1$. Since p and q do not divide $p - 1$, and $P \trianglelefteq G$, then $C_G(P) = G$, so $P \leq Z(G)$ and g commutes with h . It follows that $|gh| = pq$, so $G = \langle g, h \rangle \cong Z_{pq}$.

- If $p \mid q - 1$, then $n_p = q$ is possible. Let $Q \in \text{Syl}_q(S_q)$, and note that $|N_{S_q}(Q)| = q(q - 1)$. Cauchy's Theorem gives $P \leq N_{S_q}(Q)$ with $|P| = p$. Then PQ has order pq by [Corollary 3.15](#), and since $C_{S_q}(Q) = Q$ by [Section 4.3](#), PQ is non-abelian.

✓ **EXAMPLE:** Let $|G| = 30$. We claim this has a normal subgroup isomorphic to Z_{15} . Let $P \in \text{Syl}_5(G)$ and $Q \in \text{Syl}_3(G)$. If $P \trianglelefteq G$ or $Q \trianglelefteq G$, then [Corollary 3.15](#) shows $|PQ| = 15$, hence $PQ \trianglelefteq G$ since $|G : PQ| = 2$. However, both P and Q are characteristic in PQ by the previous example (they are unique subgroups of their orders), so $P \trianglelefteq G$ and $Q \trianglelefteq G$.

Suppose neither group is normal. Then $n_5 = 6$ and $n_3 = 10$. Note that distinct Sylow 3- and 5-subgroups intersect trivially, hence G has at least $1 + 6(5 - 1) + 10(3 - 1) = 44$ elements, contradicting $|G| = 30$. Hence, at least one of P and Q (and thus both) is normal.

✓ **EXAMPLE:** Let $|G| = 12$. We claim $n_3(G) = 1$, or $G \cong A_4$. Suppose $n_3 \neq 1$, and let $P \in \text{Syl}_3(G)$. Then $n_3 = 4$ and P has trivial intersection with its conjugates, hence there are 8 elements of order 3 in G . Moreover, $|G : N_G(P)| = 4$, hence $N_G(P) = P$. Let G act on $\text{Syl}_3(G)$ by conjugation, hence we have an afforded homomorphism

$$\varphi : G \rightarrow S_4$$

with kernel K . Note that K normalizes all Sylow 3-subgroups of G , hence $K \leq N_G(P) = P$. If $|K| = 3$, then $K = P$. Recall kernels are normal, hence $P \trianglelefteq G$, a contradiction. If $|K| = 1$, then φ is injective, so $G \cong \varphi(G) \leq S_4$. Since G contains the 8 elements of order 3 in S_4 , then $\varphi(G)$ contains all 3-cycles in S_4 , so $\varphi(G) = A_4$. Hence, $G \cong A_4$.

✓ **EXAMPLE:** Let $|G| = p^2q$ for distinct primes p, q . We show G has a normal Sylow subgroup. Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. If $p > q$, then $n_p \equiv 1 \pmod p$ and $n_p \mid q$, so $n_p = 1$, hence $P \trianglelefteq G$.

If $p < q$, note that $n_q = 1$ implies $Q \trianglelefteq G$, so suppose $n_q \neq 1$. Since $n_q \mid p^2$, then we have $n_q = p$ or p^2 . Clearly, $n_q \neq p$ since $p < q$, so $n_q = p^2$, and put $n_q = 1 + kq$ for $k \in \mathbb{Z}^+$. Then

$$kq = p^2 - 1 = (p - 1)(p + 1).$$

It follows that q divides either $p - 1$ or $p + 1$. Since $p < q$, the former cannot be true, hence $q \mid p + 1$. This forces $q = p + 1$, so $p = 2$ and $q = 3$. In this case, $n_q = 4$, hence we may use the previous example.

♠ **PROPOSITION 4.21: ORDER 60 AND SIMPLICITY**

If $|G| = 60$ and $n_5(G) > 1$, then G is simple.

Proof. Suppose $H \trianglelefteq G$ is nontrivial and proper. Note that $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 12$, hence $n_5 = 6$. Let $P \in \text{Syl}_5(G)$ with $|N_G(P)| = 10$. If 5 divides $|H|$, then H contains a Sylow 5-subgroup of G . Normality of H implies H contains all 6 Sylow 5-subgroups of G , hence $|H| \geq 6(5 - 1) = 24$. Since $|H|$ divides 60, then $|H| = 30$, but a previous example shows H contains a normal Sylow 5-subgroup.

If $|H| = 6$ or 12, then H contains a normal, hence characteristic Sylow 3-subgroup of G , which would imply normality of the Sylow 3-subgroup in G . We may replace H with a subgroup of order 2, 3, or 4. Let $\bar{G} = G/H$ with $|\bar{G}| = 30, 20$, or 15. We know $\bar{G}$ has a normal Sylow 5-subgroup $\bar{P}$ by previous results. Let H_1 be the preimage of $\bar{P}$. Then $H_1 \trianglelefteq G$ with $H_1 \neq G$, and 5 divides $|H_1|$. This contradicts the previous case, so G is simple. ■

♣ **COROLLARY 4.22: A_5 IS SIMPLE**

Proof. Note $\langle (1\ 2\ 3\ 4\ 5) \rangle, \langle (1\ 3\ 2\ 4\ 5) \rangle \in \text{Syl}_5(A_5)$. **Proposition 4.21** makes the result immediate. ■

♠ **PROPOSITION 4.23: CLASSIFYING SIMPLE GROUPS OF ORDER 60**

If $|G| = 60$ and is simple, then $G \cong A_5$.

Proof. Observe that $n_2 = 3, 5$, or 15. Let $P \in \text{Syl}_2(G)$, $N = N_G(P)$, and $|G : N| = n_2$. Observe that for some $H < G$, we claim $|G : H| < 5$. If it were, then **Theorem 4.3** gives some $K \leq H$ such that $G/K \cong S_4, S_3$, or S_2 . Simplicity of G implies $K = 1$. However, 60 does not divide 24, 6, or 2. In particular, $n_2 \neq 3$.

Suppose $n_2 = 5$. Then G acts on $\text{Syl}_2(G)$ with an afforded homomorphism $\varphi : G \rightarrow S_5$. Since G is simple, then $\ker \varphi = 1$, so $G \cong \varphi(G) \leq S_5$. If $G \not\leq A_5$, then $S_5 = GA_5$, and $G \cap A_5 \trianglelefteq G$ with $|G : G \cap A_5| = 2$, contradicting simplicity of G . Hence, $G \leq A_5$, and since $|A_5| = 60$, then $G = A_5$.

If $n_2 = 15$ and each $P \in \text{Syl}_2(G)$ was pairwise disjoint, then G has at least $15(4 - 1) = 45$ elements of order 2. Moreover, $n_5 = 6$ gives at least $6(5 - 1) = 24$ elements of order 5, so G has at least $45 + 24 + 1 = 70$ elements, a contradiction. Hence, there exist distinct $P_1, P_2 \in \text{Syl}_2(G)$ such that $|P_1 \cap P_2| = 2$. Put $N' = N_G(P_1 \cap P_2)$. Note that P_1 and P_2 are both subgroups of N' since they are abelian, and simplicity of G implies $N' \neq G$ since $P_1 \cap P_2$ cannot be normal in G . Since N' cannot have index 3 or 1, it must have index 5, and $|N'| = 12$. But we may use the preceding example to show that G collapses to A_5 , which contradicts the fact that $n_2(A_5) = 5$, by Exercise 4.5.31. Hence, $n_2 \neq 15$. ■

4.6 The Simplicity of A_n

♥ THEOREM 4.24: SIMPLICITY OF A_n

A_n is simple for all $n \geq 5$.

Proof. Inducting on n , we have the case for $n = 5$. Assume $n \geq 6$, and let $G = A_n$. Let $H \trianglelefteq G$ be nontrivial and proper. Let G_i be the stabilizer of i of G on the set $\{1, 2, \dots, n\}$ by the natural action. Then $G_i \cong A_{n-1}$ is simple by the inductive hypothesis.

Let $\sigma \in H$ with $\sigma \neq \text{id}$, but $\sigma(i) = i$ for some i . Then $\sigma \in H \cap G_i \trianglelefteq G_i$, we have $H \cap G_i = G_i$ by simplicity of G_i , or $G_i \leq H$. We know by Exercise 4.1.2 that $\tau G_i \tau^{-1} = G_{\tau(i)}$ for every $\tau \in G$, hence $G_{\tau(i)} \leq H$ for all $\tau \in G$. Since G acts transitively on $\{1, 2, \dots, n\}$, then $G_i \leq H$ for all i . Note that each $\alpha \in A_n$ is a product of $2r$ cycles, hence

$$\alpha = \alpha_1 \alpha_2 \cdots \alpha_r,$$

where α_j is a product of 2 transpositions. Since $n \geq 6$, then α_j fixes some i , hence $\alpha_j \in G_i \leq H$. It follows that $\alpha \in H$, so $H = G$, a contradiction. Hence, no nonidentity element of H fixes any i .

Suppose $\sigma_1, \sigma_2 \in H$ with $\sigma_1(i) = \sigma_2(i)$ for some i . Then $\sigma_2^{-1}\sigma_1(i) = i$, so $\sigma_2^{-1}\sigma_1 = \text{id}$ by the previous paragraph, hence $\sigma_1 = \sigma_2$. Now suppose $\sigma \in H$ has cycle decomposition that contains a 3-cycle, like

$$\sigma = (a_1 \ a_2 \ a_3 \cdots) \cdots$$

Suppose $\rho \in G$ such that $\rho(a_1) = a_1, \rho(a_2) = a_2$, but $\rho(a_3) \neq a_3$. Then

$$\sigma' = \rho \sigma \rho^{-1} = (a_1 \ a_2 \ \rho(a_3) \cdots) \cdots$$

so that σ and σ' are distinct elements of H that agree on a_1 and a_2 , contradicting the previous paragraph. Hence, no element of H has a 3-cycle in its cycle decomposition.

Consider nonidentity $\sigma \in H$ with cycle decomposition

$$\sigma = (a_1 \ a_2)(a_3 \ a_4)(a_5 \ a_6) \cdots$$

Let $\rho = (a_1 a_2)(a_3 a_5) \in G$. Then

$$\sigma' = \rho \sigma \rho^{-1} = (a_1 \ a_2)(a_5 \ a_4)(a_3 \ a_6) \cdots$$

so that σ and σ' are distinct elements of H that agree on a_1 and a_2 , again contradicting the previous argument. It follows that H cannot contain any nonidentity element, so $H = 1$, a contradiction. ■

5 Chapter 5: Direct and Semidirect Products and Abelian Groups

5.1 Direct Products

◆ **DEFINITION:** The **direct product** of groups $G_1, G_2, \dots, G_n$ with operations $\star_1, \star_2, \dots, \star_n$ is the Cartesian product $G_1 \times G_2 \times \dots \times G_n$ with the operation $\star$ defined by

$$(g_1, g_2, \dots, g_n) \star (h_1, h_2, \dots, h_n) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots, g_n \star_n h_n).$$

We may extend this to a direct product of infinitely many groups $G_1, G_2, \dots$ with operation $\star_1, \star_2, \dots$ whose elements are sequences $(g_1, g_2, \dots)$.

♠ **PROPOSITION 5.1: DIRECT GROUP IS A GROUP**

The direct product of groups $G_1, \dots, G_n$ is a group with order $|G_1| |G_2| \dots |G_n|$.

Proof. To show $G = G_1 \times G_2 \times \dots \times G_n$ is a group follows from the fact that each G_i is a group. The identity of G is $(1_1, 1_2, \dots, 1_n)$ where $1_i \in G_i$ is the identity. The inverse of $(g_1, g_2, \dots, g_n) \in G$ is $(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$ where $g_i^{-1} \in G_i$ is the inverse of g_i . The order of G is clear. ■

♠ **PROPOSITION 5.2: PROJECTIONS ONTO DIRECT GROUPS**

Let $G_1, \dots, G_n$ be groups with their direct product G .

(i) Fix some i . Then

$$G_i \cong \{(1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\} \leq G.$$

Moreover, identifying that subgroup with G_i , we have $G_i \trianglelefteq G$, and

$$G/G_i \cong G_1 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n.$$

(ii) Define $\pi_i : G \rightarrow G_i$ by $\pi_i(g_1, \dots, g_n) = g_i$. Then π_i is a surjective homomorphism with kernel

$$\ker(\pi_i) = G_1 \times \dots \times G_{i-1} \times 1 \times G_{i+1} \times \dots \times G_n.$$

(iii) If $g \in G_i$ and $h \in G_j$ for $i \neq j$, then $gh = hg$ in G .

Proof.

(i) Consider the homomorphism

$$\varphi : G \rightarrow G_1 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n, \quad \varphi(g_1, \dots, g_n) = (g_1, \dots, g_{i-1}, g_{i+1}, \dots, g_n).$$

It is easy to see $\ker \varphi = \{(1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$ and φ is surjective. By the First Isomorphism Theorem, (i) is fully proved.

(ii) Trivial.

(iii) The nonidentity parts of g and h are in different coordinates, so they commute. ■

► **NOTE:** For $g_i \in G_i$ and $k \in \mathbb{Z}$, we have $(g_1 g_2 \dots g_n)^k = g_1^k g_2^k \dots g_n^k$ by Proposition 5.2 (iii). Hence,

$$|g_1 g_2 \dots g_n| = \text{lcm}(|g_1|, |g_2|, \dots, |g_n|).$$

◆ **DEFINITION:** The **elementary abelian** group of order p^n for prime p is the group

$$E_{p^n} = \underbrace{Z_p \times Z_p \times \dots \times Z_p}_{n \text{ times}}.$$

5.2 The Fundamental Theorem of Finitely Generated Abelian Groups

◆ **DEFINITION:** G is **finitely generated** if some finite $A \subset G$ exists such that $G = \langle A \rangle$.

◆ **DEFINITION:** The **free abelian group of rank r** is the direct product of r copies of $\mathbb{Z}$, denoted $\mathbb{Z}^r$, with $\mathbb{Z}^0 = 1$. Moreover, $\mathbb{Z}^r = \langle A \rangle$, where $A = \{e_1, e_2, \dots, e_r\}$ and each e_i is the standard basis vector with 1 in the i -th coordinate and 0 in all other coordinates.

♥ THEOREM 5.3: FUNDAMENTAL THEOREM OF FINITELY GENERATED ABELIAN GROUPS

Let G be a finitely generated abelian group. Then

(i)

$$G \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_s},$$

for integers $r, n_1, n_2, \dots, n_s$ satisfying:

- (a) $r \geq 0, n_j \geq 2$ for all j , and
- (b) $n_{i+1} \mid n_i$ for $1 \leq i \leq s-1$.

(ii) The expression in (i) is unique: if $G \cong \mathbb{Z}^u \times Z_{m_1} \times Z_{m_2} \times \cdots \times Z_{m_v}$ for integers $u, m_1, m_2, \dots, m_v$ satisfying the same conditions as in (i), then $r = u, s = v$, and $n_i = m_i$ for all i .

Proof. Derived in ?? as a consequence of a more general classification theorem. An alternate proof for finite groups is given at the end of ??. ■

◆ **DEFINITION:** With respect to Theorem 5.3:

- The number r is the **free rank** or **Betti number** of G ,
- the integers n_i are the **invariant factors** of G , and
- the expression of G is the **invariant factor decomposition** of G .

◆ **DEFINITION:** If G is finite abelian with invariant factors $n_1, n_2, \dots, n_s$, then G is of **type** $(n_1, n_2, \dots, n_s)$.

► **NOTE:** For some G with type $(n_1, n_2, \dots, n_s)$, then all $n_i \mid n_1$. In particular, square free orders imply that G is cyclic, as in Corollary 5.4:

♣ COROLLARY 5.4: SQUAREFREE ORDER IS UNIQUE ABELIAN GROUP

If n is squarefree, then the only abelian group of order n is the cyclic group Z_n , up to isomorphism.

Proof. Squarefree order implies G has type $n_1 = n$ and $n_j = 1$ for all other j , hence $G \cong Z_n$. ■

✓ **EXAMPLE:** We have $180 = 2^2 \cdot 3^2 \cdot 5$. Note that $2 \cdot 3 \cdot 5 \mid n_1$, so the only choices for n_1 are $2^2 \cdot 3^2 \cdot 5$, $2^2 \cdot 3 \cdot 5$, $2 \cdot 3^2 \cdot 5$, and $2 \cdot 3 \cdot 5$. For each choice, we construct potential n_2 with $n_2 \mid n_1$ and $n_1 n_2 \mid n$. Similarly, we may choose n_3 with $n_3 \mid n_2$ and $n_1 n_2 n_3 \mid n$. We get the following list of sequences:

- $n_1 = 2^2 \cdot 3^2 \cdot 5, n_2 = 1, n_3 = 1$ corresponds to Z_{180} .
- $n_1 = 2 \cdot 3^2 \cdot 5, n_2 = 2, n_3 = 1$ corresponds to $Z_{90} \times Z_2$.
- $n_1 = 2^2 \cdot 3 \cdot 5, n_2 = 3, n_3 = 1$ corresponds to $Z_{60} \times Z_3$.
- $n_1 = 2 \cdot 3 \cdot 5, n_2 = 6, n_3 = 1$ corresponds to $Z_{30} \times Z_6$.

♥ THEOREM 5.5: PRIMARY DECOMPOSITION THEOREM FOR FINITE ABELIAN GROUPS

Let G be abelian with order n that has prime factorization $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$. Then

- (i) $G \cong H_1 \times H_2 \times \cdots \times H_k$ where $|H_i| = p_i^{\alpha_i}$,
- (ii) for each $H \in \{H_1, H_2, \dots, H_k\}$ with $|H| = p^\alpha$, then

$$H \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_m}}$$

where $\beta_1, \beta_2, \dots, \beta_m$ form a partition of α such that $\beta_{i+1} \leq \beta_i$ for $1 \leq i \leq m-1$, and

- (iii) the decomposition in (i) and (ii) is unique: if $G \cong K_1 \times K_2 \times \cdots \times K_r$ with $|K_i| = p_i^{\alpha_i}$ for all i , then $H_i \cong K_i$, and they have the same invariant factors.

Proof. Proven later. ■

◆ **DEFINITION:** With respect to Theorem 5.5:

- The integers $p_i^{\beta_j}$ are the **elementary divisors** of G , and
- the expression of G in (ii) is the **elementary divisor decomposition** of G .

► **NOTE:** If G is a p -group, say $|G| = p^\alpha$, then the number of nonisomorphic abelian groups of order p^α is the number of partitions of α . Hence, the number of nonisomorphic abelian groups of order $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ with q_i representing the number of partitions of α_i is $q_1 q_2 \cdots q_k$.

✓ **EXAMPLE:** Suppose $n = 1800 = 2^3 \cdot 3^2 \cdot 5^2$. Then we have

Order p^β	Partitions of β	Abelian Groups
2^3	3; 2, 1; 1, 1, 1	$Z_8, Z_4 \times Z_2, Z_2 \times Z_2 \times Z_2$
3^2	2; 1, 1	$Z_9, Z_3 \times Z_3$
5^2	2; 1, 1	$Z_{25}, Z_5 \times Z_5$

Then we have a total of 12 potential isomorphism types by taking all possible direct products from each abelian group of order 2^3 , 3^2 , and 5^2 :

$$\begin{array}{ll}
 Z_8 \times Z_9 \times Z_{25} & Z_4 \times Z_2 \times Z_3 \times Z_3 \times Z_{25} \\
 Z_8 \times Z_9 \times Z_5 \times Z_5 & Z_4 \times Z_2 \times Z_3 \times Z_3 \times Z_5 \times Z_5 \\
 Z_8 \times Z_3 \times Z_3 \times Z_{25} & Z_2 \times Z_2 \times Z_2 \times Z_9 \times Z_{25} \\
 Z_8 \times Z_3 \times Z_3 \times Z_5 \times Z_5 & Z_2 \times Z_2 \times Z_2 \times Z_9 \times Z_5 \times Z_5 \\
 Z_4 \times Z_2 \times Z_9 \times Z_{25} & Z_2 \times Z_2 \times Z_2 \times Z_3 \times Z_3 \times Z_{25} \\
 Z_4 \times Z_2 \times Z_9 \times Z_5 \times Z_5 & Z_2 \times Z_2 \times Z_2 \times Z_3 \times Z_3 \times Z_5 \times Z_5
 \end{array}$$

◆ **PROPOSITION 5.6: CYCLIC DECOMPOSITION CRITERION**

Let $m, n \in \mathbb{Z}^+$.

- (i) $Z_m \times Z_n \cong Z_{mn}$ if and only if $(m, n) = 1$, and
- (ii) if $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$, then $Z_n \cong Z_{p_1^{\alpha_1}} \times Z_{p_2^{\alpha_2}} \times \cdots \times Z_{p_k^{\alpha_k}}$.

Proof.

- (i) Put $Z_m = \langle g \rangle$, $Z_n = \langle h \rangle$, and $\ell = \text{lcm}(m, n)$. Recall from Section 0.2 that $\text{lcm}(m, n) = mn$ if and only if $(m, n) = 1$. Let $g^s h^t \in Z_m \times Z_n$. Then $(g^s h^t)^\ell = g^{s\ell} h^{t\ell} = 1$. If $(m, n) \neq 1$, every element would be strictly less than mn , hence $Z_m \times Z_n \not\cong Z_{mn}$.

If $(m, n) = 1$ instead, then $|gh| = mn$. The result is clear.

- (ii) Follows from (i) using induction on k . ■

► **NOTE:** Let G be abelian of type $(n_1, n_2, \dots, n_k)$. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_m^{\alpha_m} = n_1 n_2 \cdots n_k$. Put each n_i as

$$n_i = p_1^{\beta_{i1}} p_2^{\beta_{i2}} \cdots p_m^{\beta_{im}}$$

where $\beta_{ij} \geq 0$ for all i, j . Then by Proposition 5.6, we have

$$Z_{n_i} \cong Z_{p_1^{\beta_{i1}}} \times Z_{p_2^{\beta_{i2}}} \times \cdots \times Z_{p_m^{\beta_{im}}}.$$

Then the elementary divisors of G are the integers $p_j^{\beta_{ij}}$ for $1 \leq i \leq k$ and $1 \leq j \leq m$ such that $\beta_{ij} > 0$.

We may also collect all $Z_{p_j^{\beta_{ij}}}$ for $1 \leq i \leq k$ to obtain the Sylow p_j -subgroup of G .

✓ **EXAMPLE:** If $|G| = 1800 = 2^3 \cdot 3^2 \cdot 5^2$ of type $(30, 30, 2)$, observe that $Z_{30} \cong Z_2 \times Z_3 \times Z_5$. Then

$$G \cong Z_{30} \times Z_{30} \times Z_2 \cong Z_2 \times Z_3 \times Z_5 \times Z_2 \times Z_3 \times Z_5 \times Z_2,$$

and $P \in \text{Syl}_2(G)$ is isomorphic to $Z_2 \times Z_2 \times Z_2$, $Q \in \text{Syl}_3(G)$ is isomorphic to $Z_3 \times Z_3$, and $R \in \text{Syl}_5(G)$ is isomorphic to $Z_5 \times Z_5$. Hence, the elementary divisors of G are 2, 2, 2, 3, 3, 5, 5.

► **NOTE:** If G is a direct product of cyclic groups, the above process may be performed to obtain the elementary divisors of G .

✓ **EXAMPLE:** Consider the following calculations:

- (i) $G = Z_6 \times Z_{15} \cong Z_2 \times Z_3 \times Z_3 \times Z_5$ has elementary divisors 2, 3, 3, 5.
- (ii) $G = Z_{21} \times Z_6 \times Z_{15} \cong Z_3 \times Z_7 \times Z_2 \times Z_3 \times Z_3 \times Z_5$ has elementary divisors 2, 3, 3, 3, 5, 7.
- (iii) $G = Z_{22} \times Z_{42} \cong Z_2 \times Z_{11} \times Z_2 \times Z_3 \times Z_7$ has elementary divisors 2, 2, 3, 7, 11.

✓ **EXAMPLE:** Let G be an abelian group of order $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ with a given list of elementary divisors. To find the invariant factors of G , we follow the following steps:

- (i) Group the elementary divisors that are powers of the same prime together. We then obtain k lists of integers, one for each prime p_j .
- (ii) Arrange the integers in nondecreasing order in each list.
- (iii) Among the k lists, suppose that the longest (the one with the most terms) consists of t integers. If a list has fewer than t integers, then we append 1's to the end of the list until it has t integers. We then have k lists of t integers.
- (iv) For each $i \in \{1, 2, \dots, t\}$, we multiply the i -th integers from each of the k lists together to obtain n_i . We then have a list of t integers $n_1, n_2, \dots, n_t$.

Note that the ordering of these lists is structured in such a way to ensure divisibility of the invariant factors. As an example, take G with elementary divisors 2, 3, 2, 25, 3, 2. We then regroup the elementary divisors into lists of powers of the same prime:

$p = 2$	$p = 3$	$p = 5$
2	3	25
2	3	1
2	1	1

We then multiply the integers in each row together to obtain the invariant factors $n_1 = 2 \cdot 3 \cdot 25$, $n_2 = 2 \cdot 3 \cdot 1$, and $n_3 = 2 \cdot 1 \cdot 1$. Hence, the invariant factor decomposition of G is

$$G \cong Z_{150} \times Z_6 \times Z_2$$

Refer to the previous list of the 12 abelian groups of order 1800 to see that this is indeed one of the groups:

Z_{1800}	$Z_{300} \times Z_6$
$Z_{360} \times Z_5$	$Z_{60} \times Z_{30}$
$Z_{600} \times Z_3$	$Z_{450} \times Z_2 \times Z_2$
$Z_{120} \times Z_{15}$	$Z_{90} \times Z_{10} \times Z_2$
$Z_{900} \times Z_2$	$Z_{150} \times Z_6 \times Z_2$
$Z_{180} \times Z_{10}$	$Z_{30} \times Z_{30} \times Z_2$

5.3 Table of Groups of Small Order

◆ **DEFINITION:** The group $Z_3 \rtimes Z_4$ is the group of order 12 with presentation

$$\langle x, y \mid x^3 = y^4 = 1, xyx^{-1} = y^2 \rangle,$$

and $\langle y \rangle \in \text{Syl}_3(Z_3 \rtimes Z_4)$ is inverted by x acting by conjugation.

◆ **DEFINITION:** The group $(Z_3 \times Z_3) \rtimes Z_2$ is the group of order 18 with presentation

$$\langle x, y, z \mid x^2 = y^3 = z^3 = 1, yz = zy, x^{-1}yx = y^{-1}, x^{-1}zx = z^{-1} \rangle,$$

and $\langle y, z \rangle \in \text{Syl}_3((Z_3 \times Z_3) \rtimes Z_2)$ is inverted by x acting by conjugation.

◆ **DEFINITION:** The group $Z_5 \rtimes Z_4$ is the group of order 20 with presentation

$$\langle x, y \mid x^4 = y^5 = 1, x^{-1}yx = y^{-1} \rangle,$$

and $\langle y \rangle \in \text{Syl}_5(Z_5 \rtimes Z_4)$ is inverted by x acting by conjugation.

◆ **DEFINITION:** The **Frobenius group** of order 20 is the group F_{20} with presentation

$$\langle x, y \mid x^4 = y^5 = 1, x^{-1}yx = y^2 \rangle,$$

and $\langle y \rangle \in \text{Syl}_5(F_{20})$ is inverted by x acting by conjugation. This appears as the normalizer of a Sylow 5-subgroup of S_5 , namely:

$$F_{20} = \langle (2 \ 3 \ 5 \ 4), (1 \ 2 \ 3 \ 4 \ 5) \rangle.$$

► **NOTE:** The following table lists all abelian groups of order n for $1 \leq n \leq 20$ up to isomorphism: first column is order, second column is the number of nonisomorphic groups of that order, third column is the list of abelian groups of that order, and fourth column is the list of nonabelian groups of that order.

Remark: the listed nonabelian groups of order 16 are from my own lists, which Dummit and Foote do not provide.

n	Number of Groups	Abelian Groups	Nonabelian Groups
1	1	Z_1	–
2	1	Z_2	–
3	1	Z_3	–
4	2	Z_4, V_4	–
5	1	Z_5	–
6	2	Z_6	S_3
7	1	Z_7	–
8	3	$Z_8, Z_4 \times Z_2, Z_2 \times Z_2 \times Z_2$	D_8, Q_8
9	2	$Z_9, Z_3 \times Z_3$	–
10	2	Z_{10}	D_{10}
11	1	Z_{11}	–
12	5	$Z_{12}, Z_6 \times Z_2$	$A_4, D_{12}, Z_3 \rtimes Z_4$
13	1	Z_{13}	–
14	2	$Z_{14}, Z_7 \times Z_2$	D_{14}
15	1	Z_{15}	–
16	14	$Z_{16}, Z_8 \times Z_2, Z_4 \times Z_4$ $Z_4 \times V_4, Z_2 \times Z_2 \times Z_2 \times Z_2$	$Z_4 \rtimes V_4, Z_4 \rtimes Z_4, M, D_{16}, QD_{16}, Q_{16}$ $D_8 \rtimes Z_2, Q_8 \rtimes Z_2, D_8 * Z_4$
17	1	Z_{17}	–
18	5	$Z_{18}, Z_6 \times Z_3$	$D_{18}, S_3 \times Z_3, (Z_3 \times Z_3) \rtimes Z_2$
19	1	Z_{19}	–
20	5	$Z_{20}, Z_{10} \times Z_2$	$D_{20}, Z_5 \rtimes Z_4, F_{20}$

5.4 Recognizing Direct Products

◆ **DEFINITION:** Let $g, h \in G$ and $H, K \subseteq G$ be nonempty.

- Define $[g, h] = g^{-1}h^{-1}gh$, the **commutator** of g and h .
- Define $[H, K] = \langle [h, k] \mid h \in H, k \in K \rangle$, the subgroup generated by commutators of elements of H and K .
- Define $G' = \langle [g, h] \mid g, h \in G \rangle$, the **commutator subgroup** of G .

♠ **PROPOSITION 5.7: PROPERTIES OF COMMUTATORS**

Let $g, h \in G$ and $H \leq G$. Then

- $gh = hg[g, h]$,
- $H \trianglelefteq G$ if and only if $[H, G] \leq H$,
- $\sigma[g, h] = [\sigma(g), \sigma(h)]$ for any $\sigma \in \text{Aut}(G)$, G' char G , and G/G' is abelian.
- G/G' is the largest abelian quotient of G in that if $H \trianglelefteq G$ and G/H is abelian, then $G' \leq H$. Conversely, $G' \leq H$ implies $H \trianglelefteq G$, and G/H is abelian.
- If $\varphi : G \rightarrow A$ is a homomorphism into A abelian, then φ factors through G' , i.e., $G' \leq \ker \varphi$, and the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\quad} & G/G' \\ & \searrow \varphi & \downarrow \\ & & A \end{array}$$

Proof.

- Clear.
- $H \trianglelefteq G \iff g^{-1}hg \in H \iff h^{-1}g^{-1}hg \in H \iff [h, g] \in H$ for all $g \in G$ and $h \in H$.
- For $\sigma \in \text{Aut}(G)$, then

$$\sigma[g, h] = \sigma(g^{-1}h^{-1}gh) = \sigma(g)^{-1}\sigma(h)^{-1}\sigma(g)\sigma(h) = [\sigma(g), \sigma(h)].$$

Then $\sigma[g, h] \in G'$. Invertibility of σ implies $G' \subseteq \sigma(G')$, and hence $G' = \sigma(G')$. Then G' char G . Finally, for any $g, h \in G$, we have by (i) that

$$(gG')(hG') = ghG' = hg[g, h]G' = hgG' = (hG')(gG'),$$

and hence G/G' is abelian.

- If $H \trianglelefteq G$ and G/H is abelian, then for $g, h \in G$, we have

$$1H = (gH)^{-1}(hH)^{-1}(gH)(hH) = [g, h]H,$$

and hence $[g, h] \in H$.

Conversely, $G' \leq H$ and G/G' abelian by (iii) implies every subgroup of G/G' is normal, and hence $H/H' \trianglelefteq G/G'$. By the Lattice Isomorphism Theorem, $H \trianglelefteq G$. By the Third Isomorphism Theorem, then

$$G/H \cong (G/G')/(H/G'),$$

and hence G/H is abelian.

- Clear from (iv). ■

✓ **EXAMPLE:** A group G is abelian if and only if $G' = 1$. To see this, simply note that abelian G implies $[g, h] = 1$ for every $g, h \in G$. The converse is clear from assuming trivial G' .

✓ **EXAMPLE:** Let D_{2n} have usual representation. Note that $[r, s] = r^{-2}$, hence $\langle r^{-2} \rangle = \langle r^2 \rangle \leq D'_{2n}$. Moreover, $\langle r^2 \rangle \trianglelefteq D_{2n}$, since $sr^2s^{-1} = r^{-2} \in \langle r^2 \rangle$. Note that $|\bar{s}| = |\bar{r}| = 2$, hence $\bar{s}\bar{r} = \bar{r}\bar{s}$, and $D_{2n}/\langle r^2 \rangle$ is abelian. By Proposition 5.7(iv), we have $D'_{2n} \leq \langle r^2 \rangle$, and hence $D'_{2n} = \langle r^2 \rangle$. If n is odd, then $\langle r^2 \rangle = \langle r \rangle$, whereas n even implies $\langle r^2 \rangle$ is of index 2 in $\langle r \rangle$. Hence, D'_{2n} is of index 2 or 4 in D_{2n} whether n is odd or even, respectively.

♠ **PROPOSITION 5.8: COUNTING FACTORIZATION IN PRODUCT SUBGROUP**

Let $H, K \leq G$. Then the number of distinct factorizations of $g \in HK$ as $g = hk$ with $h \in H$ and $k \in K$ is $|H \cap K|$. In particular, $H \cap K = 1$ implies each $g \in HK$ has unique factorization as $g = hk$ with $h \in H$ and $k \in K$.

Proof. Let $g = h_0 k_0 \in HK$ for some $h_0 \in H, k_0 \in K$. Then $g \in h_0 K$. If $g = hk$ for some $h \in H, k \in K$, then $g \in hK$, hence $hK = h_0 K$. This implies $h^{-1}h_0 \in K$, and since $h^{-1}h_0 \in H$, we have $h^{-1}h_0 \in H \cap K$. Set $h_0 = hx$ for some $x \in H \cap K$. Since $g = hk = h_0 k_0$, we know $hk = hxk_0$, hence $k = xk_0$. Then every factorization of g has the form $g = (h_0 x^{-1})(xk_0)$ for $x \in H \cap K$. Conversely, for any $x \in H \cap K$, we have $g = (h_0 x^{-1})(xk_0)$, and hence the number of distinct factorizations of g is $|H \cap K|$. ■

♥ **THEOREM 5.9: DIRECT PRODUCT RECOGNITION THEOREM**

Let $H, K \leq G$ such that

- (i) $H \trianglelefteq G$ and $K \trianglelefteq G$, and
- (ii) $H \cap K = 1$.

Then $HK \cong H \times K$.

Proof. We know $HK \leq G$ by Corollary 3.15. Let $h \in H$ and $k \in K$. Observe $[h, k] \in H \cap K = 1$, hence $hk = kh$. By Proposition 5.8, every $g \in HK$ has unique factorization as $g = hk$ with $h \in H$ and $k \in K$. Define $\varphi : H \times K \rightarrow HK$ by $\varphi(h, k) = hk$. This is a bijection by the uniqueness of factorization. Moreover, for $(h_1, k_1), (h_2, k_2) \in H \times K$, we have

$$\varphi((h_1, k_1)(h_2, k_2)) = \varphi(h_1 h_2, k_1 k_2) = h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi(h_1, k_1) \varphi(h_2, k_2),$$

and hence φ is a homomorphism. Therefore, $HK \cong H \times K$. ■

♦ **DEFINITION:** Let $H \trianglelefteq G$ and $K \trianglelefteq G$ with $H \cap K = 1$. Then HK is the **internal direct product** of H and K . We call $H \times K$ the **external direct product** of H and K .

✓ **EXAMPLE:** Let n be positive odd. We claim $D_{4n} \cong D_{2n} \times Z_2$. Let D_{4n} have its usual presentation. Put $H = \langle s, r^2 \rangle$ and $K = \langle r^n \rangle$. It is clear that H has index 2 in D_{4n} , since we may just associate it with the regular n -gon inscribed in the regular $2n$ -gon, hence $H \trianglelefteq D_{4n}$. Moreover, $sr^n s^{-1} = r^{-n} = r^n$, hence s centralizes r^n , and $K \trianglelefteq D_{4n}$. Since r^2 always has odd order, then $r^2 \notin K$, hence $H \cap K$ is trivial. Then Theorem 5.9 finishes the proof.

✓ **EXAMPLE:** Let $I \subseteq \{1, 2, \dots, n\}$. Let G be the setwise stabilizer of I in S_n , i.e.,

$$G = \{\sigma \in S_n \mid \sigma(I) = I\}.$$

Let $J = \{1, 2, \dots, n\} - I$. Note $\sigma(J) = J$ for $\sigma \in G$. Let P_I and P_J be the pointwise stabilizers of I and J respectively:

$$P_I = \{\sigma \in S_n \mid \sigma(i) = i \text{ for all } i \in I\}, \quad P_J = \{\sigma \in S_n \mid \sigma(j) = j \text{ for all } j \in J\}.$$

By definition, $\sigma \in G$ setwise stabilizes both I and J , i.e., decomposes into cycles $\sigma_I \sigma_J$, where $\sigma_I \in P_I$ and $\sigma_J \in P_J$. It follows that $G = P_I P_J \cong P_I \times P_J$ by Theorem 5.9. Lastly, we may extend permutations of J into permutations in S_n by letting them act as the identity on I , which are precisely the permutations in H . Similarly, the permutations of K are those that act as the identity on J , hence,

$$P_I \cong S_J, \quad P_J \cong S_I, \quad G \cong S_{|J|} \times S_{n-|J|}.$$

✓ **EXAMPLE:** Building off the previous example, suppose $\tau \in S_n$ such that τ pointwise stabilizes I . Let $C = C_{S_n}(\tau)$, and observe that τ stabilizes both I and J , by Exercise 4.3.18. Then $C \cong T \leq P_J \times P_I$. Observe that $\tau \in P_I$. Then each $\gamma \in C$ is written in the form $\gamma_I \gamma_J$ for $\gamma_I \in P_I$ and $\gamma_J \in P_J$. If $\rho \in P_J$, then $\tau \rho = \rho \tau$. It follows that C contains all ρ , or that $P_J \leq C$. Hence, every element of C has form $\gamma_I \gamma_J$ such that $\gamma_J \in P_J$ is arbitrary, and γ_I commutes with τ :

$$C_{S_n}(\tau) = C_H(\tau) \times K \cong C_{S_J}(\tau) \times S_I.$$

Moreover, if τ is an ℓ -cycle, then

$$C_{S_n}(\tau) = \langle \tau \rangle \times S_{n-\ell}.$$

5.5 Semidirect Products

♥ THEOREM 5.10: SEMIDIRECT PRODUCT CONSTRUCTION

Let H, K be groups and $\varphi : K \rightarrow \text{Aut}(H)$ be a homomorphism. Let $\cdot$ be the action of K on H determined by φ . Let $G = H \rtimes K$, and define $\star$ on G as:

$$(h_1, k_1) \star (h_2, k_2) = (h_1 k_1 \cdot h_2, k_1 k_2).$$

- (i) G is a group with order $|G| = |H||K|$.
- (ii) The sets $G_H = \{(h, 1) \mid h \in H\}$ and $G_K = \{(1, k) \mid k \in K\}$ are subgroups of G . The maps $h \mapsto (h, 1)$ and $k \mapsto (1, k)$ are isomorphisms, and $H \cong G_H, K \cong G_K$.

Identifying H and K with their copies in G , we have

- (iii) $H \trianglelefteq G$,
- (iv) $H \cap K = 1$,
- (v) for $h \in H$ and $k \in K$, $khk^{-1} = k \cdot h = \varphi(k)h$.

Proof.

- (i) Trivial, via group actions. Note that $1 = (1, 1)$, and $(h, k)^{-1} = (k^{-1} \cdot h^{-1}, k^{-1})$.
- (ii) Trivial to see $G_H, G_K \leq G$. The maps are clear isomorphisms.
- (iii) For $(h_0, 1) \in H$ and $(h, k) \in G$, then

$$(h, k)(h_0, 1)(h, k)^{-1} = (hh_0, k)(k^{-1} \cdot h^{-1}, k^{-1}) = (hh_0 h^{-1}, 1) \in H,$$

hence $H \trianglelefteq G$.

- (iv) Clear.
- (v) Use a similar process from (iii) to deduce (v). ■

♦ **DEFINITION:** Let H and K be groups with homomorphism $\varphi : K \rightarrow \text{Aut}(H)$. Then G in Theorem 5.10 is the **semidirect product** of H and K with respect to φ , denoted as $H \rtimes_{\varphi} K$.

♠ PROPOSITION 5.11: REDUCTION OF SEMIDIRECT PRODUCT TO DIRECT PRODUCT

Let H, K be groups with homomorphism $\varphi : K \rightarrow \text{Aut}(H)$. Then the following are equivalent:

- (i) $\text{id} : H \rtimes K \rightarrow H \times K$ is an isomorphism,
- (ii) φ is trivial, and
- (iii) $K \trianglelefteq H \rtimes K$.

Proof.

- By definition, id is a homomorphism when $(h_1, k_1)(h_2, k_2) = (h_1 k_1 \cdot h_2, k_1 k_2) = (h_1 h_2, k_1 k_2)$ for all $h_i \in H$ and $k_i \in K$. Equating factors, we get $h_1 k_1 \cdot h_2 = h_1 h_2$, which is equivalent to φ being trivial.
- $\varphi = \text{id}$ implies $khk^{-1} = k \cdot h = h$, hence $kh = hk$, and H commutes with K . From that, $H \trianglelefteq K$ and $K \trianglelefteq H$, hence $K \trianglelefteq H \rtimes K$.
- $K \trianglelefteq H \rtimes K$ implies $[h, k] \in H \cap K = 1$ for all $h \in H$ and $k \in K$, hence $kh = hk$, and φ is trivial. It follows $(h_1, k_1)(h_2, k_2) = (h_1 k_1 \cdot h_2, k_1 k_2) = (h_1 h_2, k_1 k_2)$ for all $h_i \in H$ and $k_i \in K$, and hence id is an isomorphism. ■

✓ **EXAMPLE:** Let H be abelian and $K = \langle k \rangle \cong \mathbb{Z}_2$. Define $\varphi : K \rightarrow \text{Aut}(H)$ by mapping k to the inversion map, hence $k \cdot h = h^{-1}$ for all $h \in H$. Then $H \leq G$ has index 2, and $khk^{-1} = h^{-1}$ for all $h \in H$. If $H = \mathbb{Z}_n$, then $G \cong D_{2n}$, and if $H = \mathbb{Z}$, then $G \cong D_{\infty}$.

◆ **DEFINITION:** The **generalized quaternion group** of order 2^{n+1} is the group $Q_{2^{n+1}}$ with presentation

$$\langle h, k \mid h^{2^n} = k^4 = 1, k^{-1}hk = h^{-1}, h^{2^{n-1}} = k^2 \rangle.$$

✓ **EXAMPLE:** To construct $Q_{2^{n+1}}$, let $H = \langle h \rangle \cong Z_{2^n}$ and $K = \langle k \rangle \cong Z_4$ with $khk^{-1} = h^{-1}$ in G . Note $k^2 \in Z(G)$ since $k \cdot (k \cdot h) = k \cdot h^{-1} = h$. Let $\langle z \rangle \leq H$ be the unique subgroup of order 2, where $z = h^{2^{n-1}}$. Since $z = z^{-1}$, then $kzk^{-1} = z$, hence $z \in Z(G)$. Then $k^2z \in Z(G)$, hence $\langle k^2z \rangle \leq G$. Note that k^2z has order 2, hence $\bar{G} = G/\langle k^2z \rangle$ has order 2^{n+1} . We may now identify $\bar{k}^2$ and $\bar{h}^{2^{n-1}}$ in $\bar{G}$.

For $n = 2$, we have that $\bar{k} = \bar{h} = 4$, $\bar{k}^2 = \bar{h}^2$, and $\bar{k}$ inverts $\bar{h}$. Hence, $\bar{G} \cong Q_8$. For general $n \geq 3$, we claim that $Z(\bar{G}) = \langle \bar{k}^2 \rangle$: We first note that $k^2z = \bar{1}$, hence $\bar{k}^2 = \bar{z} = \bar{h}^{2^{n-1}}$. From this, it is clear that $\langle \bar{k}^2 \rangle \leq Z(\bar{G})$. Conversely, if $\bar{x} \in Z(\bar{G})$, then

- If $\bar{x} = \bar{h}^i$, then observe that $\bar{k}^{-1}\bar{h}^i\bar{k} = \bar{h}^{-i} = \bar{h}^i$ implies $i = 0$ or $i = 2^{n-1}$, hence $\bar{x} \in \langle \bar{k}^2 \rangle$.
- If $\bar{x} = \bar{h}^i\bar{k}$, then $\bar{h}^{-1}\bar{x}\bar{h} = \bar{x}$ implies $\bar{h}^{-1}\bar{h}^i\bar{k}\bar{h} = \bar{h}^i\bar{k}$. Using the inversion relation, the left-hand side is $\bar{h}^{-1}\bar{h}^i\bar{h}^{-1}\bar{k} = \bar{h}^{i-2}\bar{k}$, hence $\bar{h}^2 = \bar{1}$, which is impossible. Therefore, $\bar{x} \notin Z(\bar{G})$.

► **NOTE:** For $g \in G$ infinite and $H \leq G$, then $gHg^{-1} \leq H$ does not imply $gHg^{-1} = H$. For example, let $H = (\mathbb{Q}, +)$ and $K = \langle k \rangle \cong \mathbb{Z}$ with $k \cdot h = 2h$ for all $h \in H$, and consider $\mathbb{Z} \leq H$. Observe that $k\mathbb{Z}k^{-1} = 2\mathbb{Z} \leq \mathbb{Z}$, but $k^{-1}\mathbb{Z}k \not\leq \mathbb{Z}$.

◆ **DEFINITION:** Let H and $K = \text{Aut}(H)$ with $\varphi = \text{id}$. Then $H \rtimes K$ is the **holomorph** of H , denoted $\text{Hol}(H)$.

✓ **EXAMPLE:** Some examples of holomorphs are as follows:

- (i) $\text{Hol}(Z_2 \times Z_2) \cong S_4$.
- (ii) If $|G| = n$ with $\pi : G \rightarrow S_n$ the left regular representation, then $N_{S_n}(\pi(G)) \cong \text{Hol}(G)$. In particular, because the left regular representation of a generator of Z_n is an n cycle, we have

$$N_{S_n}(\langle (1\ 2 \cdots n) \rangle) \cong \text{Hol}(Z_n) = Z_n \rtimes \text{Aut}(Z_n),$$

where $|Z_n \rtimes \text{Aut}(Z_n)| = n\varphi(n)$.

✓ **EXAMPLE:** Let p, q be primes with $p < q$. Put $H = Z_q$ and $K = Z_p$. A previous example showed $p \nmid q - 1$ implies that pq is cyclic. We may also see that $\varphi : Z_p \rightarrow \text{Aut}(Z_q)$ must be trivial as well. If we assume that $p \mid q - 1$, then there exists nontrivial $\varphi : Z_p \rightarrow \text{Aut}(Z_q)$. Then the group $G = H \rtimes K$ has order pq with K not normal in G , hence G is nonabelian.

✓ **EXAMPLE:** Let p be prime. There are two nonabelian nonisomorphic groups of order p^3 , as follows:

- (i) Let $H = Z_p \times Z_p$ and $K = Z_p$. Then $\text{Aut}(H) = \text{GL}_2(\mathbb{F}_p)$ with order $(p^2 - 1)(p^2 - p)$. Since p divides $|\text{Aut}(H)|$, there exists $\varphi \in \text{Aut}(H)$ with order p , hence nontrivial. Then $H \rtimes K$ is nonabelian with order p^3 . In particular, let $H = \langle x \rangle \times \langle y \rangle$ and $K = \langle z \rangle$. Then

$$z \cdot x = xy, \quad z \cdot y = y,$$

defining the action of K on H . If we represent the action of z on x and y as vectors, we may consider them as a nonsingular linear transformation:

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \in \text{GL}_2(\mathbb{F}_p),$$

and $H \rtimes K$ has presentation

$$\langle x, y, z \mid x^p = y^p = z^p = 1, xy = yx, zxz^{-1} = xy, zyz^{-1} = y \rangle.$$

- (ii) Let $H = Z_{p^2}$ and $K = Z_p$. $|\text{Aut}(H)| = p(p - 1)$ then admits a nontrivial homomorphism $\varphi : K \rightarrow \text{Aut}(H)$, hence $H \rtimes K$ is nonabelian of order p^3 . If $H = \langle h \rangle$ and $K = \langle k \rangle$, then

$$k \cdot h = h^{1+p}.$$

$H \rtimes K$ will have presentation

$$\langle h, k \mid h^{p^2} = k^p = 1, khk^{-1} = h^{1+p} \rangle.$$

It is clear that Examples (i) and (ii) are not isomorphic, since (ii) has elements of order p^2 , namely the element h , while (i) does not.

◆ **DEFINITION:** The **Heisenberg group** over $\mathbb{Z}/p\mathbb{Z}$ is the group above generated by $\langle z, x \rangle$.

✓ **EXAMPLE:** Let $H = Q_8 \times (Z_2 \times Z_2) = \langle i, j \rangle \times (\langle x \rangle \times \langle y \rangle)$ and $K = \langle z \rangle \cong Z_3$. Define the map in $\text{Aut}(H)$ as

$$i \mapsto j, \quad j \mapsto ij, \quad x \mapsto y, \quad y \mapsto xy,$$

which has order 3 (simply check that $\varphi^3(h) = h$ for all $h \in H$). Equipped with some $\varphi : K \rightarrow \text{Aut}(H)$ by mapping z to the specified automorphism, we get

$$z \cdot i = j, \quad z \cdot j = ij, \quad z \cdot x = y, \quad z \cdot y = xy.$$

Then $H \rtimes K$ is non-abelian with order 96 such that $i^2x \in G'$, but $i^2x \neq [g, h]$ for any $g, h \in G$.

♥ **THEOREM 5.12: SEMIDIRECT PRODUCT RECOGNITION THEOREM**

Let $H, K \leq G$ such that

- (i) $H \trianglelefteq G$, and
- (ii) $H \cap K = 1$.

Let $\varphi : K \rightarrow \text{Aut}(H)$ map $k \in K$ to left conjugation by k on H . Then $HK \cong H \rtimes K$. If $G = HK$ such that H and K satisfy (i) and (ii), then G is the semidirect product of H and K .

Proof. $H \trianglelefteq G$ implies $HK \leq G$. We know by [Proposition 5.8](#) that each $g \in HK$ is of the form hk for $h \in H$ and $k \in K$. The map $hk \mapsto (h, k)$ is a set bijection, and proving that it is a homomorphism has been previously done. Thus, the result is proved. ■

◆ **DEFINITION:** Let $H \leq G$. The **complement** of H is some $K \leq G$ such that $G = HK$ and $H \cap K = 1$.

► **NOTE:** There is a certain process of finding out the isomorphism classes of a given group of order n :

- (i) Show the group has subgroups H, K that satisfy [Theorem 5.12](#), and $G = HK$.
- (ii) Find isomorphism types of H and K .
- (iii) For each (H, K) in (ii), find all possible $\varphi : K \rightarrow \text{Aut}(H)$.
- (iv) For each (H, K, φ) in (iii), construct $H \rtimes_{\varphi} K$. Among these triples, determine which ones are isomorphic.

✓ **EXAMPLE:** Let $|G| = pq$ for primes p, q . Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. We know $G \cong Q \rtimes P$ for $\varphi : P \rightarrow \text{Aut}(Q)$. Note that P and Q are cyclic. Moreover, $\text{Aut}(Q)$ is cyclic with order $q - 1$. If $p \nmid q - 1$, then the only homomorphism from P to $\text{Aut}(Q)$ is trivial, hence $G \cong P \times Q \cong Z_{pq}$.

If $p \mid q - 1$, put $P = \langle a \rangle$. Cyclicity of $\text{Aut}(Q)$ implies a unique subgroup of order p , say $\langle \alpha \rangle$. Then any $\varphi : P \rightarrow \text{Aut}(Q)$ maps a to powers of α , or $\varphi_i(a) = \alpha^i$ for $0 \leq i \leq p - 1$. Note that $Q \rtimes_{\varphi_0} P \cong Q \times P$ since φ_0 is trivial. Each φ_i gives rise to a non-abelian group G_i of order pq . To see that $G_i \cong G_j$ for some $i \neq j$, note that there is $a_i \in P$ such that $\varphi_i(a_i) = \alpha$. Hence, $G_i \cong G_1$ for all $i \neq 0$. Then there are $p - 1$ non-abelian groups of order pq up to isomorphism, and one abelian group of order pq .

✓ **EXAMPLE:** Recall every group of order 30 has a subgroup H of order 15. Since $15 = 3 \cdot 5$, then $H \cong Z_{15}$ and it has index 2 in G . Moreover, there exists $K \in \text{Syl}_2(G)$. Then $G = HK$ with $H \cap K = 1$. Then there exists $\varphi : K \rightarrow \text{Aut}(H)$ such that $G \cong H \rtimes K$. Now,

$$\text{Aut}(Z_{15}) \cong (\mathbb{Z}/15\mathbb{Z})^{\times} \cong Z_4 \times Z_2,$$

where we know that $H = \langle x \rangle \times \langle y \rangle \cong Z_5 \times Z_3$. Since $\langle x \rangle \text{ char } H$ and $\langle y \rangle \text{ char } H$, we have

$$\text{Aut}(H) \cong \text{Aut}(Z_5) \times \text{Aut}(Z_3) \cong Z_4 \times Z_2.$$

Then $\text{Aut}(H)$ contains exactly 3 elements of order 2, which are the maps:

$$\begin{cases} x \mapsto x \\ y \mapsto y^{-1} \end{cases} \quad \begin{cases} x \mapsto x^{-1} \\ y \mapsto y \end{cases} \quad \begin{cases} x \mapsto x^{-1} \\ y \mapsto y^{-1} \end{cases}$$